ERC
EUROPEAN ROVER CHALLENGE

AGH UNIVERSITY OF KRAKOW, POLAND
4–6TH OF SEPTEMBER 2026

COMPETITION RULES

2026

roverchallenge.eu

1. Record of Changes

Revision	Date	Section	Description
1	20.02.2026	All	Public document version – initial issue.
2	26.03.2026	§6.4 Delivery of Documentation Preliminary Report Template	Science Planning Report Delivery Date changed to be aligned with Appendix 1 – Schedule. Template updated (Financial Part), Rev. 2 added.
3	19.07.2026	§4.3 Personal Data and Intellectual Property – Storage and Use §6.4 Delivery of Documentation Appendix 1 – Schedule Requirements Final Report Template	Paragraph's name updated. External partners and LLM tools added as exceptions. Dates updated (meetings and deliverables) in order to reflect the current state. Dates updated (meetings and deliverables) in order to reflect the current state. REQ-OPS-130 was clarified – no impact on design. Typo corrected. Final RF Form is not included in the 35 pages limit.

2. Lists

2.1 Contents

2.1 Figures

2.2 Tables

3. General Information

3.1 European Rover Challenge – Introduction

The European Rover Challenge (ERC) is an integrated programme working towards technological developments with space exploration utilised as the leading theme. The goal of the ERC is to become a standardised test trial and benchmark for planetary robotic activities, coupled with strong professional career development platform. The ERC consists of an engineering project where university teams build robots to compete on an extra-terrestrial inspired arena performing tasks based on international roadmaps for space robotics. This means that the competition tasks present the same level of problems as industry drivers for space robotics which have been devised for future decades. Perhaps most importantly, whilst set against the background of a competition, the ERC is a continuous mentoring effort which is intended to educate the next generation of multidisciplinary engineers, boost innovation in research and business, and popularise STEM (Science, Technology, Engineering and Mathematics) advancements, all conceptually rooted in future space exploration.

The European Rover Challenge is owned and coordinated by the European Space Foundation, organized in cooperation with a group of independent experts who make up the steering and jury boards. These rules and tasks are prepared by the Polish Space Professionals Association's and its experts.

3.2 Schedule and Venue

ERC is a venue independent, year-round programme. For information about the current ERC edition venue, please follow our updates on the challenge website. The official schedule can be found in the Appendix 1 – Schedule.

ERC 2026 is organized in Kraków on the premises of AGH University of Science and Technology.

3.3 Information Channels and Contacts

The Challenge's website address: www.roverchallenge.eu.

Teams Contact Point's email address: teams@roverchallenge.eu.

The official communication channel for challenge announcements is the list of email addresses provided by Teams during registration process. In addition, Slack tool will be used to assure the daily-based contact with the Teams participating in ERC. All Team Members will be duly invited by the Organizer to register on the ERC Slack group.

3.4 Rules

Rules of the ERC 2026 are included in the following complementary documents:

- Rules with all appendices to this document (e.g. templates, matrix of compliance)
- Presentations for Teams (these will be organized for the teams to explain rules and requirements as well as to allow participation in the Q&A session)
- Update Report #3 (details about the final arrangements of the tasks)

All the mentioned above documents can be found on www.roverchallenge.eu (in the competitor's zone tab).

3.5 Milestones

ERC is divided into milestones. This section provides the list and rules of each milestone the teams will approach during the ERC preparation and performance. The milestones and schedule is attached in the Appendix 1 – Schedule.

Milestone	Milestone Name	Deliverables	Delivered By
KO	Kick-Off: Registration End	Registration Forms	Teams
PDR-1	Preliminary Design Review: Step 1	Preliminary Design Report Preliminary Radio Frequency Form	Teams
PDR-2	Preliminary Design Review: Step 2	Video	Teams
QUALIFICATION			
CDR-1	Critical Design Review: Step 1	Final Design Report	Teams
CDR-2	Critical Design Review: Step 2	Final Radio Frequency Form Science Planning Report	Teams
ERC FINALS			
TRR	Test Readiness Review	Team presence ag the briefing is required	Teams
PTR	Post Test Review	Participation in the ERC tasks on the Mars Yard	Teams
FP/FR	Final Presentation/Review	Participation in the Presentation Task	Teams
THE END OF THE ERC			

Table 1: ERC Milestones

Content of the abovementioned documentation is provided in the templates.

In addition, **Update Reports** will be published by the organizer. Those milestones do not need any deliverables from the Teams but are intended to clarify requirements and provide additional information to the Teams to proceed with and finalise the rover's design. The milestone will be delivered in a form of report sent to the Teams contact point's email address.

- **Update Report #1** (update of the Rules including scoring, remaining documents templates)
- **Update Report #2** (clarifications from the organizer, remaining documents templates)
- **Update Report #3** (so called Environmental Report that includes data required for the final competition and Science Planning Report – 3D model of the Mars Yard and relevant description, drone photos, locations of the points on the Mars Yard incl. starting points, navigation waypoints, sampling locations etc.)

3.6 Awards and Recognitions

The first three best teams will be presented an award. Multiple other awards are planned to recognise excellence in different parts of the programme and competitions. The form of the awards will be specified on the challenge website. The Organizer may also allow awards funded by third parties. Third-party award funders must have the Organizer's approval.

The following awards are foreseen:

- **1st place**
- **2nd place**
- **3rd place**
- **Best team awards for each task accomplished (showing exceptional performance or approach, not necessarily based on the sum of points scored), i.e.**
  - ✓ **Science – Scientific Exploration**
  - ✓ **Science – Sampling**
  - ✓ **Science – Astro-Bio**
  - ✓ **Navigation – Traverse**
  - ✓ **Navigation – Droning**
  - ✓ **Maintenance**
  - ✓ **Probing**
  - ✓ **Presentation**
- It is considered that other type of awards may be granted, however decision on this depends on other factors, e.g. budget, sponsorships, fair play attitude of the teams etc.

By taking part in the ERC, teams agree to place a promotional sticker on their rover (max. size of the sticker: 10x10cm).

Number of commemorative medals for each awarded Team is limited to 30 (medals are not applicable for the best team awards for tasks accomplished). In case there are more than 30 members of the team present in Poland the additional ones over 30 may receive certificates only.

It is foreseen that the winning rovers (i.e. places 1st-3rd) will get the commemorative sticker placed on it by the organizer for each place.

Digital certificates of participation in the ERC will be given only to the participants who actively participated in development of the rover for the competition's final (not necessarily were present in Poland, however members of teams who were present in Poland will receive additional line in the certificate confirming their participation in the final event). The team is responsible for defining who has actively participated in these activities (at the registration and to be confirmed after completion of the ERC, the number of team members will be verified – situation that a team has grown by significant number of 'active' members will be considered as unrealistic). System for registration will be provided by the organizer. Moreover, issuing certificates for team members that have not provided filled GDPR forms will not be possible.

No certificates will be given to the teams who do not appear in Poland for the ERC final (i.e. qualification without attendance is not sufficient).

3.7 Qualification

3.7.1 Qualification to the Finals

The ERC finals are planned for a limited number of teams. At the moment up to 20 teams are considered by the organizer, however this number may be changed until announcement of the final qualification lists. Together with the jury board, the organizer will choose which of the registered teams will be invited to compete in the challenge. The choice will be made based on the documentation delivered by teams (described in the documentation section and in the templates). The organizer will announce qualifying teams before another deadline given in the schedule.

3.7.2 Wild Card

The organizer reserves the right to **qualify up to 5 teams based on subjective assessment of their performance** (e.g. despite of worse performance in the delivered reports which is the usual case for the new teams, the delivered video shows exceptional performance of their rover and drone. Therefore, only 15 out of 20 teams (in the worst case) can be qualified based on the achieved scoring.

3.8 Organizer's Disclaimer

Teams take full responsibility for any damages, accidents or unsettling events caused by their hardware as well as by the members of the team. It is especially critical in case of drone flying – it is allowed only in the droning cage that will be provided by the organizer. **Any flights performed outside of the drone cage are teams' responsibility (consider local droning rules and obtaining additional aircraft damage insurance – Aero Casco if you plan to perform any test flights outside the droning cage).** Teams are obliged to follow all safety and good conduct rules specified by the organizers. Any breaches of safety rules and requirements will result in the disqualification of the team from the entire competition.

3.9 Changes to the Competition Rules

The organizer retains the right to extend the deadline for the submission of documents and provide essential but inevitable changes to the competition rules. However, any changes introduced cannot concern key issues for the rover's design. All changes will be announced in advance and provided on the challenge's website.

3.10 Deadline Extension

The organizer has the right to extend the deadline for submission of documents. All deadline extensions will be announced in advance and provide details on the challenge website. Deadline extensions are not granted based on a single team's requests and will be provided only in case of a clear need based on the current circumstances that might have significantly affected time available for teams to deliver the required documentation.

3.11 Q&A

Answers to any challenge related questions that arise will be provided on the challenge's website, SLACK group or during the Q&A meetings with teams. If you have questions, please contact the challenge contact point.

The organizer will provide 'European Rover Challenge Questions & Answers' meetings as a part of the competition rules. All arrangements contained therein are ultimately binding – even if they change the competition rules. The Q&A meetings will be announced in advance and presentations from those will be provided on the challenge's website (in the Rules catalogue).

3.12 Challenge Scoring Issues

All issues with scoring during the challenge will be resolved solely by the independent jury (i.e. challenge judges or head of jury). Teams cannot appeal to any other party.

3.13 Organizational Issues

Organizational issues, including team eligibility, challenge organization and execution of jury decisions will be resolved by the organizer.

3.14 General Challenge Issues

In case any conflict related to the challenge is encountered, the organizer's decision will be considered as final and binding.

3.15 Disqualification

The organizer may disqualify a team in the event of a serious breach of the rules, safety regulations or fair play.

3.16 Cancellation of Event

The organizer reserves the right to cancel the ERC finals in the event of circumstances preventing its safe organization. In case of event cancellation, the organizer will decide on the alternative approach and present it to the teams affected by the decision.

3.17 Organizer's Responsibility

The organizer's civil liability is limited solely to the responsibility for organizing a mass event in accordance with the Polish law and local regulations.

3.18 Copyright

The organizer retains all copyright to the competition rules, especially the description of the tasks. No alterations or additions to the competition rules can be made and their sale is expressly forbidden. The rules can only be used or copied for the ERC-connected activities (e.g. registration process).

4. Teams

4.1 Team Members

75% of a team must be comprised of higher education students and recent graduates (i.e. max two years after receiving master's or PhD degree): undergraduate and masters-degree level students (with no limitations) and PhD students. It is highly recommended that teams cooperate with specialists from different institutions, but students must prepare and sign all the required documentation themselves.

A team may consist of students from more than one higher education institution. An institution may also be affiliated with more than one team. Team membership is exclusive – each person can only be a member of one team.

In case the abovementioned requirement is not fulfilled, please contact the organizer to confirm if your team can register for the competition.

4.2 Registration

The Team registration shall be done by filling and sending the form provided by the organizer (link: https://forms.gle/do7Fqv9DBKQTkaJv7). It shall include e.g.:

- The name of the institution with which the team is affiliated (if the team is affiliated with more than one institution, please list all the names, in descending order of involvement)
- Team name
- Rover name (may be the same as the team's name)
- Short planning description (in case the Team takes part in both on-site and remote formulas)
- The approximate number of team members who plan on coming to the challenge (i.e. appear on site)
- Team contact point: the contact's name and surname, telephone number and email address **(remember that this email address will be used as the main contact. No information from the organizer will be delivered to any other address – please refer to the "Comments on the contact point")**.
- University team coordinator/supervisor and key members: name and surname, telephone number and email address
- Project website address and Facebook fan-page (if applicable)
- The following declaration in English:

  'By sending this application and registering the team to the European Rover Challenge each team member fully accepts all terms and provisions of the ERC rules and all final decisions of the European Rover Challenge organizer.'

- To register the team to participate in the competition, complete the application form, sign GDPR form (print and scan separate for each team member mentioned with name and surname in the form) and attach it in the registration form. In case the team qualifies to the ERC finals, GDPR forms for all team members have to be delivered (scan of the signed by hand form or certified signature – NOT the ones provided in various PDF reader applications), otherwise issuing certificates for the team members that have not delivered GDPR forms will not be possible

In case of any issues with attaching the requested files in the registration form, please send these to the following address: teams@roverchallenge.eu. Use your contact point's email address (see 'General comments on the CONTACT POINT' below) and confirm with us that we have received your registration.

General comments on the CONTACT POINT:

- All teams are requested to create alias address teamname@xx.yy (where xx and yy are not defined and can be selected by the teams).
- It is advised that the mentioned alias shall resend all the received messages to all members of the team in order to minimise chance to omit any of the information received from the organizer.
- It has to be highlighted that messages from the organizer will be send only to the defined contact point's email address (teamname@xx.yy, see above). The organizer is responsible only for sending all the details to the contact point's email address. Teams are responsible to check the provided contact point's emails on a regular basis in order not to miss any important communication from the organizer.
- No response will be provided to the email addresses other than the contact point. These emails will be considered as spam.

The organizer will try to respond to all messages posted on the ERC SLACK group, however short time of response cannot be guaranteed in this case. The primary contact will be Q&A sessions. Those will be held 1-2 weeks after every update of the Rules and before the important ERC milestones (delivery of documentation, briefings before ERC finals).

4.3 Personal Data and Intellectual Property – Storage and Use

Team members agree to their personal data, the documentation delivered as well as other promotional materials and visuals being stored and processed in the organizer's computer systems for the purpose of the ERC programme.

On the other hand, the organizer will keep all technical documentation confidential and will not publish or disclose it to any third parties without prior approval from the team's representatives. The sole exception to this is the challenge's jury team – technical documentation will be disclosed to judges for scoring and mentoring purposes only. This exception is applicable also to additional tasks organized by ERC's external partners, such as ERC side tasks organized by external entities and use of AI/LLM tools that may be used to confirm scores provided by judges and originality of the provided documentation.

The team members also give the organizer, parties designated by the organizer and the audience, the right to disclose and publish any photos, videos or other visuals, their names and surnames, identifiable pictures of themselves and any other persons, as well as pictures of machines, devices, and equipment in any and all of the available formats, by any and every known method, in any and every known medium.

Teams grant permission to the organizer to use promotional materials and visuals (e.g. photos and videos), as well as any additional photos, videos, portraits, documents, interviews, and other materials resulting from participation in the challenge (using the name of the participant or not) on all media, in any language, anywhere in the world, in any manner, for advertising and promotional purposes.

Personal data and information about team members other than their names and surnames will not be published without the prior consent of each team member.

4.4 Teams' and Members' Responsibilities

Teams and team members accept sole responsibility for securing and ensuring the safety of their equipment and luggage in the challenge's location. They release the organizer from liability in the event of damage, destruction, or theft of any property.

5. Requirements

5.1 Rover and Drone Requirements

Each rover and drone must be compliant with the requirements listed in Appendix 3 – Requirements to take part in the challenge. Special cases of non-compliance should be discussed with the organizer as soon as possible in the development process. The organizer has the right to exclude the team from field trials, especially when non-compliances are reported too late (e.g. during the finals). It is highly recommended that teams present their status of compliance with the specified requirements in a transparent way in their technical reports (i.e. Preliminary Report and Final Report).

5.2 Communication Requirements

Radio communication with the rover shall be designed for use of legally available frequencies and power levels.

- **It is expected that the maximum distance between the rover and the antenna mast would be less than 100 meters.**
- **It shall be highlighted that direct line-of-sight between the control base and rover antennas may be occluded by different forms of terrain morphology.**
- **Using standard communication such as WiFi 2.4 GHz may be difficult as there will be multiple networks available in the vicinity (university networks, organizer's networks, private networks established by people in the audience, bluetooth devices etc.). These networks may interfere with the participants' networks. Therefore, the rover and drone communication performance shall be checked before the competition in such environment. It is recommended to e.g. use 5 GHz WiFi instead of 2.4 GHz (more channels are available and less issues are expected but at the same time**
- **Judges at ERC will organize band assignment, to separate frequences between teams.**
- **In rare cases the same WiFi channel can be assigned to multiple teams.**
- **Radio channel interference is not a base for task redo.**

5.2.1 Accepted Frequencies

The following frequencies can be accepted for the ERC.

Radio Amateur Bands
Accepted bands up to 1 W signal transmitted and up 10 W EIRP.

- 144 - 146 MHz
- 430 - 440 MHz
- 1240 - 1300 MHz
- 5650 - 5850 MHz

It is highly recommended that each Team should have at least one member with an amateur radio license (CEPT class T/R 61-01). When using amateur radio bands, team should provide a scan of the license of at least one member.

2.4 GHz Band (2412 - 2472 MHz)
Only Wi-Fi communication standard is accepted (IEEE 802.11). Any other systems such as analogue video cameras or RC controllers are forbidden.

- Accepted channels: 1-13 (centre frequency 2412 MHz – 2472 MHz). WiFi channel 1 spans from 2400 MHz and the last Wi-Fi channel (14 – not accepted at the ERC) is till 2495 MHz. that range is protected and cannot be used by any other communication devices than the WiFi 802.11.
- Up to 100 mW EIRP
- Accepted standards: IEEE 802.11 b/g/n
- Rover can use only one 20 MHz channel
- SSID should be set to "ERC_TeamName".

5 GHz Low Band (5150 - 5725 MHz)
Inside the band between 5150 and 5725MHz team can use two 40 MHz channels for Wireless Access Systems: Wi-Fi and AirMax only. Up to 1 W EIRP power is permitted. Wider channel allocation (80 MHz) is possible only after acquiring a permission from the organizers by providing reasonable technical arguments (official email contact with **teams@roverchallenge.eu** is required). SSID should be set to "ERC_TeamName_(A/B)". At least one of the two allocated channels will not be a DFS channel (if possible to set by the Team on their equipment).

5 GHz High Band (5725 - 5875 MHz)
Inside the band between 5725 and 5875 MHz team **can use two 30 MHz channels for any communication system - FPV etc.**

ISM Bands
It is possible to use ISM bands within their limitations but Team must designate which rule is in compliance with European regulations (LINK). The ERC does not accept ISM bands which are not accepted in Europe (e.g. 915 MHz).
Voice communication using a 500 mW PMR licensed transceiver is allowed on following channel frequencies (MHz):

- 446,00625
- 446,01875
- 446,03125
- 446,04375
- 446,05625
- 446,06875
- 446,08125 - reserved for the organizer
- 446,09375 - reserved for the organizer

Mobile Networks
Communication via mobile networks – e.g. 4G, 5G, LTE is allowed.

Other Frequencies
Other frequencies are only allowed when a relevant license, valid in the territory of the venue, is presented by the team. Those communication channels must be described in the documentation and agreed with the organizer.

5.2.2 Other Communication Rules

- **Before the competition, rovers and ground stations must be checked and accepted by the radio communication judge during an Radio Frequency Check, ensuring that proper channels and power limits have been set. Team should come early enough before competition to make changes and tests for the communication system. When long preparation occur, time slot for competition will be missed. Team should bring 2x filled Radio Frequency Check Cards (RFCC) (see Appendix 4 – RFCC) for every task, after successful Radio Frequency Check RFCC will be signed by RF judge. These cards have to be provided to judge responsible for the task. Channels on different bands will be assigned to the teams on the RF Check just before the competition.**
- **Channels on different bands will be assigned to the teams on the RF Check just before the competition.**
- During the competition, rovers and ground stations will be randomly tested. Unauthorized changes to the RF configuration may result in immediate disqualification.
- The use of any communication channels for testing (at any time outside of the duration of the competition attempt) must be consulted with the RF judge. Testing that can be conducted without RF communication is preferred. The organizer will provide rules for the usage of RF links for the main parts of the challenge venue. Requests to limit the usage of RF links can be expected and should be respected throughout the entire duration of the event.

- For the whole duration of the challenge, the team is responsible for the legal use of the frequencies used in the venue's territory. The organizer can only help with frequency coordination and does not take responsibility for any license violations, such as exceeding RF power, frequency band or area of use.
- **It has to be highlighted that organizer is not responsible for any communication issues encountered by the teams. Nevertheless the organizer will make all the reasonable steps to assign different channels for competing teams as well as assure that other equipment does not use the same ranges of frequencies, there may be some external interferences as multiple teams can be driving in the Mars Yard or another Wi-Fi networks may be present in the vicinity of the Mars Yard.**
- In case you would miss your time slot for execution of the tasks on the Mars Yard you need to take into account that no alternative slot may be granted. In addition, no delays in starting ERC tasks (induced by participants) will be allowed in order to minimise possibility of encountering interferences between the teams (channels will be given to the teams based on their scheduled rides, multiple teams will be using Mars Yard at the same time therefore in case there is any deviations with regards to the schedule, the risk for interferences between the teams is higher.
- Team should deliver correct RF form at least two weeks before participating, to be able to receive points for RF Form.
- Difference between radio frequency form and real hardware during competition will result in negative points for the Radio Frequency Form up to –20 points for each rule violation.
- Using unauthorized radio channel parameters (frequency, power, using additional channels etc) during competition will result in negative points (-20 pts for each rule violation).

Use of drone in the Droning Sub-Task

- This task is to be performed inside the cage, therefore no regulations are applicable as drones will be used in fully enclosed and safe environment.
- However, please take into account that if you decide to fly outside the competition and outside the cage, e.g. to test something you need to follow the local PL/EU UAV flying rules – please check the local rules for use of the drones in Poland. Also check website of the ULC (Civil Aviation Authority in Poland)). **Any flights performed outside of the drone cage are teams' responsibility (consider local droning rules and obtaining additional aircraft damage insurance – Aero Casco if you plan to perform any test flights outside the droning cage).**
- The accepted frequencies are as mentioned above in the section 5.2.1. Exception: RC controllers are not forbidden in case of drones.
- Use of GPS/GNSS is allowed.
- **The drone can be used only in the Droning Sub-Task.**
- **Before the competition drone weight is measured, and the Radio Frequency Form will be checked by RF Judge Team.**

6. Deliverables

6.1 General Information

Each team must provide technical documentation for their project. The project documentation is divided into two parts called the Preliminary Report and the Final Report. Additionally, teams are required to deliver the video (showing its rover's and drone's performance) and the Science Planning Report. All should present the required content and quality of professional engineering documentation, therefore it is strongly recommended to have it reviewed by experienced engineers.

All the documents are scored and counted as a part of the challenge final points (for details, see Appendix 2 – Scoring). Scoring is designed to consider documentation as an aspect that can have a significant impact on the final results, therefore it is important to deliver all documents in the best possible quality according to the requirements listed below and on time according to the schedule (see Appendix 1 – Schedule).

6.2 Documentation

6.2.1 Preliminary Report

See the document template for more details. The template provides requirements associated with the document, e.g. maximum number of pages, font size, margins etc.

6.2.2 Final Report

See the document template for more details. The template provides requirements associated with the document, e.g. maximum number of pages, font size, margins etc.

6.2.3 Radio Frequency Form

Each team must fill in a Radio Frequency Form for every RF module used. It must be submitted via online forms together with the Preliminary and the Final Report. If these documents are not submitted in the requested form, the team will not be allowed to participate in the challenge. Radio Frequency Form can be found in Appendix 5 – RF Form.

Preliminary RF Form is used to check compliance with the ERC rules. In case of any non-compliance, the team will be requested to change communications. Final RF Form provides an ultimate rover's and drone's configuration, and it will be checked against the real configuration during the competition's final. Any differences will result in penalties.

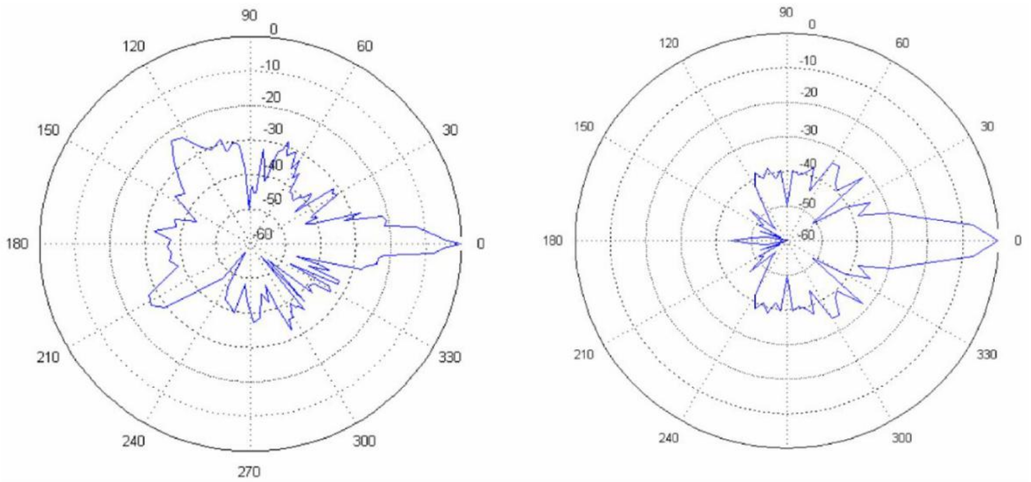

Figure 1: Example of horizontal and vertical RF antenna radiation pattern for ground station

6.2.4 Science Planning Report

Each team needs to prepare a Science Planning Report and send it before the ERC finals (deadline is provided in Appendix 1 – Schedule). Late delivery will be punished by proportionally subtracting 20% for every 24 hours after the deadline. Information on detailed requirements are provided in the description of the Science Task.

DOCUMENTATION – GENERAL COMMENTS:

- Check document requirements (margins, font size, which pages are not included in the limit etc.)! Refer to the attached templates!
- Include all the relevant information mentioned in the templates
- Try not to elaborate too much, sometimes a clear chart is better than a page full of text

6.3 Video

Each team should prepare promotional video material presenting their readiness for the competition. This deliverable must be completed and submitted by the date presented in the challenge schedule (see Appendix 1 – Schedule). The material should be uploaded to one of the popular video services (e.g. YouTube) **and made publicly available**. A relevant link must be delivered to the organizer by the specified deadline.

The video should be a maximum of 10 minutes in length and present the rover's and drone's capability to take part in the challenge, containing the following elements:

1) Introduction of team name, rover name, and the higher education institution name
2) Introduction of team and present teamwork
3) Introduction of the reasons for running in the challenge
4) Presentation of the remote control, mobility, design and safety systems for the rover
5) Presentation of the remote control, mobility, design and safety systems for the drone
6) Presentation of manipulator and other tools as well as ability to resolve the challenge tasks (manipulator, drill, gripper etc.)
7) Quality and proper visual aesthetics for a video clip and presentation skills

Refer to Appendix 2 – Scoring for more details.

In special cases, the video may be the basis to request more details concerning the team's readiness to participate in the competition. Failure to present the requisite level of readiness may influence the extent to which the team is allowed to participate in the trials.

VIDEO – GENERAL COMMENTS:

- Remember about the rover's and drone's motion sequence required in the video
- Check scoring, some of the moves shall be performed without cuts – in a single sequence recorded
- It is highly recommended to divide your video in sections covering elements 1-7. By implementing this approach, you can assure that all required elements and motions will be noticed and scored by judges. In case any of the elements appears in the wrong section it may be not counted in the final scoring.

6.4 Delivery of Documentation

Documents shall be delivered to Organiser using the following links:

No.	Event	Link	Deadline
1	**Preliminary Report**	https://forms.gle/QT7SsKjd917QEESV8	8th April 2026 23:59 CEST / EOD
	Preliminary RF Form	https://forms.gle/xkzJCRfx79fyipjX9	8th April 2026 23:59 CEST / EOD
2	**Video**	https://forms.gle/q8ybHawswrJ6H9ri7	31st May 2026 23:59 CEST / EOD
3	**Final Report**	https://forms.gle/kvLZuCoeUx5Wm7Zb9	26th July 2026 23:59 CEST / EOD
	Final RF Form	https://forms.gle/frUCWmuPhuYbsULCA	26th July 2026 23:59 CEST / EOD
4	**Science Planning Report**	https://forms.gle/kCqBdSf1HtePEQy28	23rd August 2026 23:59 CEST / EOD
5	**Droning Sub-Task Report**	https://forms.gle/SczzpirtDypA4uQb9	2nd September 2026 23:59 CEST / EOD
	Droning Sub-Task Photos	https://forms.gle/55xneJsci7uSQHfh8	up to 2.5 hours after the ride on the Mars Yard
6	**Science Exploration / Final Report**	https://forms.gle/SpvVA3vyrMACiEaD9	up to 2.5 hours after the ride on the Mars Yard
7	**Science AstroBio / Final Report**	https://forms.gle/adm2dFXY2QXMMd8HA	up to 2.5 hours after the ride on the Mars Yard
8	**Presentation File** (PDF preferred, no videos needed)	https://forms.gle/my8oZr4ymGZpLUfW8	5th September 2026 23:59 CEST / EOD

Table 2: ERC Deliverables with Links

7. ERC 2026 Tasks

7.1 General Information – the Story

PROLOGUE – MARTIAN ENVIRONMENT RECONNAISSANCE MISSION – AUTONOMOUS INTERIOR DATABASE (MERMAID)

It is year 2050. The successful execution of ARTEMIS programme has proven that people are able to build bases on the Moon. The next step is to perform the same activities on Mars. This task is far more demanding, especially due to the distance between the Earth and the Mars. It results in no possibility for sending a fast rescue and recovery missions. This is the reason why we must be sure that everything was done to assure the highest possible level safety for the first astronauts that will reach the Red Planet.

The first people are going to land on the Red Planet in the next 5 years. Nevertheless, to minimize the remaining risks a rover mission was successfully launched and landed on the surface of Mars, near the place that was selected for the first astronaut's landing. The rover descended from the lander using the customized egress system and it is ready for operations. The plan established for this mission assumes execution of the following tasks and sub-tasks:

Science Task (Sub-Tasks: Exploration, Surface/Deep Sampling, Astro-Bio)

This task consists of four sub-tasks. The purpose of this task is to recreate the typical scientific activities performed by the rover on the surface of the Mars (exploration of the landing area, collection of surface and deep samples, looking for biological markers and life). This part is performed just after the landing on the Red Planet's surface is exploration of the neighbourhood area.

Navigation Task (Sub-Tasks included: Traverse, Droning)

When all the planned Science tasks are finished the rover must find its way back to the MLMOD (Martian Lander Module) to deliver the collected samples that will be sent back to Earth. Based on the current location of the rover established with the data from satellite positioning system in the MMO (Medium Mars Orbit) the track back to the lander was uploaded to the rover's computer. Traverse must be performed in autonomous mode due to limited bandwidth and delays encountered while controlling the rover manually from the Earth.

During the traverse activities (in some of the locations on the way and due to difficult terrain) it is also required to use the drone that is part of the rover that has landed on the Mars. The main task of the drone is to fly over the surface of Mars in order to support reconnaissance activities – i.e. to detect potential obstacles impeding the movement of the rover, as well as to assess the value of the terrain in terms of future sampling activities. The autonomously moving drone will have to find a sealed probe containing Martian material that is going to be flown back to Earth for further investigation. Due to the likelihood of a sandstorm, it should also be possible to temporarily land in a safe place and then continue the mission. The drone's destination will be MLMOD.

Maintenance Task

It was an unexpected plot twist. So far everything went nominal and when the rover arrived to the MLMOD it was noticed that a device designed to analyse surface and subsurface humidity changes has failed after the recent dust storm. After unloading and setting the instrument on the ground, but before finally disconnecting it from the lander's boom, a final test was carried out. It turned out that there were problems with the instrument's power supply. The fault could not be repaired remotely. Manual intervention is required. Fortunately, one of the multi-purpose, autonomous Mars rovers was nearby. The rover can perform the necessary operations on the instrument's control panel to repair the fault. The mission control centre has already sent the instrument data to the operators and the rover. Now the success of the mission depends only on the dexterity of the rover operators and the technical capabilities and reliability of their design.

Probing Task

Three years later as part of the final preparations to the manned mission a set of 5 small and fast rovers was sent to the Mars to collect the probes that were left during the previous mission. The probes performed environmental measurements that are crucial for the upcoming manned mission.

Presentation Task

After successful repairs to the MLMOD the collected samples were loaded on the lander and sent back to Earth. Re-entry manoeuvres went nominally, the samples were successfully delivered to the surface of the Earth and collected by the mission's scientific team. Thorough analysis of the mission execution and the collected samples allowed for several scientific publications that were presented during some of the most important symposia and scientific conferences associated with the current and future trends related to exploration of the Red Planet. These research activities cleared the way for the first manned mission to Mars, that was finally launched in 2056.

All tasks and sub-tasks will be performed separately (i.e. Science Task: Exploration, Astro-Bio, Sampling; Navigation Task: Traverse, Maintenance). Each team will be assigned a time slot for execution of each task.

7.2 Field Trials

Field trials are organized as a benchmarking activity to compare the performance of teams in the resolution of several tasks. Each task presents an independent set of problems to be solved and which are related to the technologies required by future space robotics missions.

7.2.1 General

- The challenge tasks take place in front of an audience in the form of a public event
- Challenge attempts are independent. Teams are permitted to change their rover configuration between tasks. A certain amount of time will be scheduled in between tasks to allow teams to modify, repair and optimize their rovers
- The challenge jury consists of a number of specialists selected by the organizer. While judging the challenge, the jury acts independently of the organizer but adheres to the schedule provided by the organizer. In case of any unforeseen issues not specified in the competition rules, the jury board will propose a solution
- The scoring of each task is independent and summarised in Appendix 2 – Scoring.

- Excellence shown in a particular task can be rewarded with additional points or multiplication factors (see scoring details)

7.2.2 Schedule

- On the first day, teams should register at the challenge venue
- Additionally, for all teams, a warm-up day is planned for the day before the challenge. This day should be used for calibration and other preparation activities. The organizer will give each team a limited time slot within which teams are allowed to do any kind of measurements agreed with the organizer and based on the final report specifications. Some task elements which are considered too detailed may be removed for this day by the organizer. All dynamic elements may not be placed in their final locations. The organizer cannot guarantee that the challenge area and its elements will be fully ready for this day
- On the last day of the challenge, total scores are calculated, winners are announced and prizes awarded
- A detailed schedule containing the exact time window for each task will be announced by the organizer one week before the event in a preliminary version, with the final one being issued on the first day of the competition
- Each team is obliged to respect the schedule. Any request for the modification of time slots etc. may be rejected by the jury without any reason given

7.2.3 Challenge Site Details

- Each challenge task can be organized either indoors or outdoors. The outdoor challenge elements may be placed under tents. Teams can expect typical interior furnishings, buildings, industrial installations (metal pipes etc.) and natural objects (e.g. trees, bushes) in the vicinity of the challenge area
- For outdoor tasks, teams and their systems should be prepared for a range of weather conditions. Temperatures between 10°C and 30°C, wind gusts, light drizzle, moderate rain showers, strong or weak levels of sunlight are acceptable. This will be made based on the schedule, other requests and the potential impact on team's performance. In case of major weather problems, the organizer will make all reasonable efforts to reschedule/reorganise the trials within the available time slots and facilities. However, it cannot be guaranteed that all of the trials will take place or will be organized strictly in accordance with the provided specifications
- The organizer will provide each team with a workspace equipped with tables, chairs and 230V, 50Hz power socket ('type E', compatible with the 'German type F')
- The challenge location is separated from the Team area to avoid RF interference but the organizer cannot guarantee that extra precautions will not be requested to avoid disruption of the challenge attempts
- The challenge field (i.e. Mars Yard - the place where terrain dependent tasks are held) will be artificially landscaped specifically for the event. Sandy, non-cohesive soil, as well as hard, dry terrain, all at a variety of slope angles and a large number of stones and boulders of different sizes should be expected. In case of tasks which do not score locomotion aspects, a flat industrial surface (e.g. concrete) can also be expected.

7.2.4 Operations

- The aim of the challenge is to demonstrate and evaluate the performance and robustness of the proposed solutions. All tasks are designed to eliminate 'luck factor' from challenges. Therefore, teams shall present high level of readiness for each task and platforms shall be equipped with all the devices needed to solve all task elements. Rovers that are not equipped with all of the necessary elements may not be allowed to attempt a certain task
- For the reasons stated above, teams can expect dynamic elements in the task description i.e. elements that will be defined separately for each team at the beginning of the attempt (e.g. changing the start position, different positions of task elements etc.). In those cases, judges will propose a fair modifications and the team cannot influence those decisions
- Teams will control their rovers from the independent locations. The control stations locations will be set up by the organizer so that team members that are in control of the rover will not see their rover during the tasks.

- Each team has about 20-60 minutes (if a task description does not state otherwise) to complete each task. This value will be fixed by the time of the final schedule release
- Some of the tasks may be realized during night, as part of the schedule or as result of severe weather conditions during the day
- Each team must designate two observers who are allowed to follow the rover at a safe distance to ensure the safety of the machine and others around. Observers are allowed to communicate with the team from the control area only through a judge, and only in one way - from control base to observer - and solely to coordinate actions unrelated to task details like task reset, abort or unsafe event. No communication during the normal execution of the task is allowed. The observers must be able to carry the rover, but they should remain at a safe distance from the working machine and cannot interfere with any of the rover's sensors (e.g. be visible on the image from the camera) during the realization of the task attempt
- During the tasks, only judges and observers can access the Mars Yard. No manual intervention is allowed, except during events where the task rules state otherwise
- Any maintenance made by the team during tasks (any operations made by the team to the rover hardware on the field) results in a restart of the task, going back to the start line and cancellation of all previously earned points for this task (unless the ERC scoring states otherwise)
- The team can use video systems to tele-operate the rover if the task requirements do not state otherwise
- The team shall not use any voice/visual communication with the crew on the Mars Yard. Only the judge can communicate between the Mars Yard and the control station
- The rover's operator has the right to abort the task at any time by notifying the judge about it. The team will receive the points gathered to the moment of notification according to the rules of the task
- Throughout the entire event, no rover or any other part of the system may do harm or interfere with the systems of other teams. Any reports of such breaches will be investigated independently by the judges or the organizer, and any violation of this rule can lead to disqualification from the challenge
- Any erratic behaviour of the rover or team members causing damage to the task's infrastructure can result in the immediate termination of the task attempt and cancellation of all the collected points

7.3 Tasks' Definition

REMINDER:

It shall be taken into consideration that all tasks in the Mars Yard will be performed <u>separately</u> (i.e. Science Task with Collection, Navigation, Maintenance). Each team will be assigned a time slot for execution of each task. Changes of rover's configuration <u>are allowed</u> (between the tasks).

7.3.1 Science Task

Science Task consists of the following sub-tasks:

- Scientific Exploration
- Surface Sampling
- Deep Sampling
- Astro-Bio

Descriptions of these sub-tasks are provided in this section.

7.3.1.1 Scientific Exploration Sub-Task

The aim of the Science Task is to demonstrate the **understanding of performing scientific exploration in planetary geology conditions**.

The workflow of the Scientific Exploration Sub-Task is as follows:

1. Analysis of previous relevant work about the Martian region to be explored (literature review)
2. Careful analysis of provided data about the Mars Yard

3. Preparing a scientific exploration plan that considers the requirements of the data collection
4. Then, execution of the science exploration plan on the Mars Yard during the ERC field trials (Science Exploration Sub-Task)
5. Data analysis
6. And finally, delivering a final report

The Scientific Exploration Sub-Task will be divided into **three parts**.

1. **"Science: Regional Geology Analysis"** is graded as a part of the **"Final Report"** that is part of the ERC finals
2. **"Science Planning"** that is required to be delivered about a **week before the ERC finals**
3. **"Scientific Exploration"** is to be submitted up to **2,5 hours after the ride** through the Mars Yard

All reports shall be submitted as separate PDF files using the links listed in the section number 6.4. The PDF file should be signed in the following format {[name of the team]_[report_name]}. If you do not follow this naming instruction, 5% points will be deducted from your final grade. Please note that the maximal length of the answer to each question will be strictly enforced (material exceeding the allowed number of characters will not be read and assessed). The Task documentation is scored according to Appendix 2 – Scoring. **Late submissions will be penalised** (for part 1 and 2 every 24 hours means 20% points less, for the part 3 each 60 minutes delay is 20% less – penalties are proportional.

At some point after the finals, every team may request feedback with the Science Team after the ERC (delivered remotely at a mutually convenient time).

Materials provided by the ERC organizer

You will be provided with the following data based on which you will prepare your reports:

1. Detailed instruction (below)
2. An approximate location of the Mars Yard on the surface of the Red Planet. Please note that the Mars Yard does not copy any area of the real Mars but is an artificial model that represents the entire region. This information should help you to better understand the geological context of geomorphological forms present on the Mars Yard
3. Drone images of the Mars Yard along with the digital elevation model of the area that will be delivered at the latest three weeks before the finals of ERC
4. A non-compulsory workshop that will introduce various topics helpful to accomplish Science Tasks (link: https://www.youtube.com/live/QNf3-r8kX6g)
5. Support of our Science Team through the dedicated contact platform.

Required documentation:

Part 1: "Science: Regional Geology Analysis"

The goal is to select and shortly describe (in a form of a map and text) a best landing site on a pre-defined region on Mars. Selected landing site must be geologically interesting (within a rover traverse up to 30 km), possible to land there based on currently available techniques (appropriate atmospheric thickness, appropriate landing area) and safe to access by a rover (local slopes inclination <20°, lack of numerous large boulders, reasonable thermal conditions for a rover to survive winter). To prepare your analysis you should use Google Earth Pro (choose Mars as a base planet) or, if you are more ambitious, JMars. The report should consist of two parts fitting on two pages A4:

1) A **composite map** fitting on an **A4 format** and consisting of a 1) location map showing selected landing site in a global scale, 2) a regional geological map (based on an appropriate map from the Astrogeology Science Center publications) and 3) a close-up map based on CTX or Hirise photos, with a clearly marked a) landing ellipse, b) planned rover path, c) three regions of interest
2) A **report**: up to **800 words** long **technical** (why is it safe to land there) and **scientific** (why is it worth landing there – especially in reference to the three regions of interest) justification of the selected area.

The region selected for the ERC2026 is within the deepest impact basin on Mars: **Hellas Planitia (+/- 100 km from -36.409170° 89.679479°)**. You should expect an interaction between volcanic features (from nearby Hadriacus Mons) and features related to past and contemporary ice presence in the subsurface. Our Mars Yard will be roughly modelled on this geological area on Mars.

Suggested reading:

- https://pubs.usgs.gov/imap/i2694/
- https://www.sciencedirect.com/science/article/pii/S0169555X24003787?via%3Dihub
- https://www.sciencedirect.com/science/article/abs/pii/S0032063308002432
- https://www.sciencedirect.com/science/article/abs/pii/S0019103522004912?via%3Dihub

Part 2: "Science Planning"

The goal is to analyse the Mars Yard as a "landing site" and design a scientific mission in this area.

This task's documentation is based on information (drone footage and digital elevation model) provided no later than 3 weeks before the ERC finals, the report is due 1 week before ERC finals as a pdf file. Science Planning report must contain detailed information on the topics listed below.

1. Geological Map of the Mars Yard including a coherent geological interpretation of the 'landing site'.

Outline of all geological features visible on the Mars Yard with a focus on what they are, how they were formed, and what their relative ages are (assigned from the oldest to the youngest). The map should fit on an A4 format of a page.

Preparing a good geological map of the area is *the most important part of the scientific part of the Science Task*: without a good map you will not be able to obtain good grades from other tasks.

An example of a geological map: Geologic Map of Jezero Crater and the Nili Planum Region, Mars by V.Z. Sun and K.M. Stack can be found here: https://pubs.er.usgs.gov/publication/sim3464

2. An exploration plan: Based on the prepared geologic map, and the understanding of geological development of the area, you should prepare a detailed **plan of scientific exploration** and properly justify your decisions. This section should be no longer than allocated number of characters (with spaces) and include assigned number and type of figures. Longer descriptions WILL NOT BE READ, more than one (possibly composite) figure will not be taken into account.

a) 1st paragraph: stating a subject to be studied. e.g., Relative ages of Aeolian features present at the landing site. (up to 500 characters with spaces)

b) 2nd paragraph: explaining why we should know that. e.g., determining the geological history of the landing site, and establishing when (and maybe why) dunes became immobilized. (up to 1000 characters with spaces)

c) 3rd paragraph: a falsifiable hypothesis: e.g., an impact crater is younger than the dune located next to it. (up to 500 characters with spaces). Wiki has a good article on what is a good hypothesis: https://en.wikipedia.org/wiki/Hypothesis You should particularly make sure your hypothesis is fulfilling requirements of the Scientific Hypothesis as described in this article. Hypothesis should be related to the geologic analysis present in your map, and cannot be ridiculously obvious like: "is this object a rock?", "is this dune made of sand?". Every team may ask once (through a direct message through the selected communication channel e.g., slack) if your hypothesis is too ridiculously obvious)

d) 4th paragraph + one A4 figure (pdf format): <u>predictions</u> that will test the hypothesis; what is expected to be seen in the field, and how it will help the team sustain or overrule the hypothesis e.g., we will go to the slope of the dune facing the crater and examine how material ejected from the crater interacts with the dune. If we find ejected material on the dune's surface, we will know that the crater is younger than the dune → it therefore supports our hypothesis. If we will find that material ejected from the crater is covered by the dune, we will know that the dune is younger than the crater → our hypothesis will be rejected. A figure can be used to illustrate the expected results (up to 1500 characters with spaces)

e) Relevant scientific references that were <u>appropriately cited in the text</u> (do not cite Wikipedia or NASA web pages – we want to see you read papers or book chapters).

<u>Part 3: "Scientific Exploration"</u>

The goal is to:

1. Verify the hypothesis stated in the Scientific Plan.
2. Observe the surroundings for potentially scientifically interesting objects that will be placed around the Mars Yard (e.g., atypical rocks, meteorites, pieces of spacecraft, possible USO (Unexpected Standing Objects).

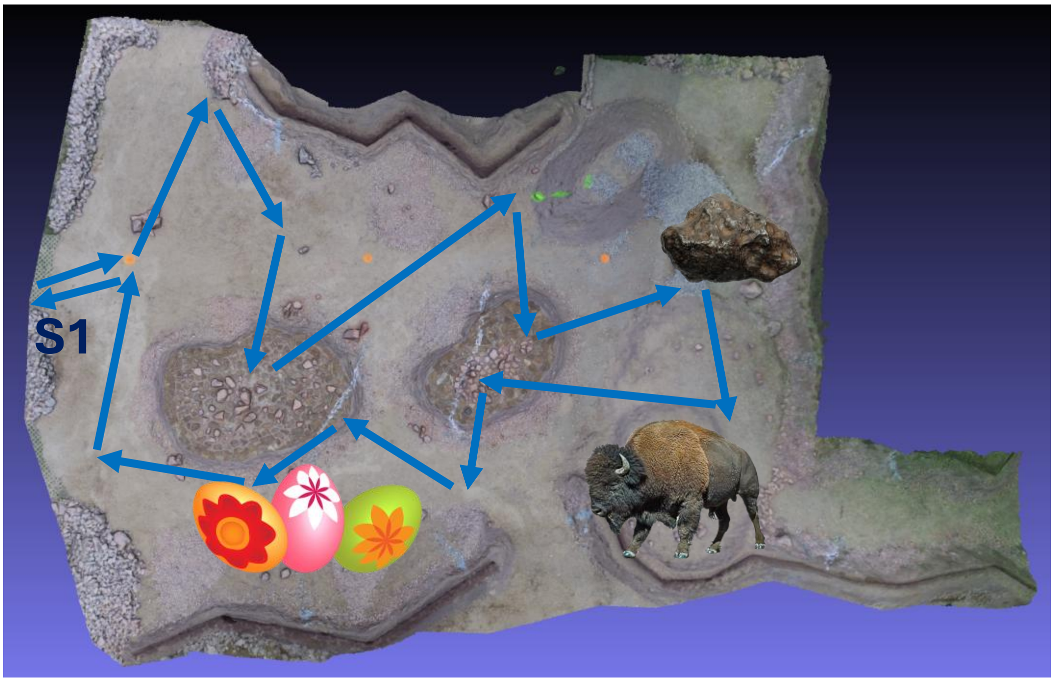

Figure 2: Example of Scientific Exploration based on ERC 2024. Start from S1 point, exploration of the Mars Yard with discovery of easter eggs and geological features, finally return to S1.

<u>Deadline for submitting the report: up to 2.5 hours after completing the traverse (refer to §6.4 for the delivery link).</u>

<u>Hypothesis verification:</u>

Aim: an attempt to verify your hypothesis described in your Scientific Plan Report. We will judge you on the quality of your data analysis against the hypothesis. It is fine to reject the hypothesis, it is also ok to determine that the data was not sufficient to properly test the hypothesis (although in this case you should explain what else would be needed to test it). Within your explanation you should include three annotated photographs from the rover showing geologic features that you discuss in the report. You should discuss how this new knowledge influences the understanding of the geology of this area and modify the geological map appropriately. You should note the location of your observations on the geological map and update geological map including all new or modified features (there will be some!!). Please make sure that the new/ changed features are clearly marked on the map.

USO: Unexpected Standing Objects (up to 3 objects): during your traverse you should identify and describe "unexpected objects" identified on the Mars Yard. When doing real science (on Earth and on other planetary bodies) usually the most important science is related to finding new, unexpected things while looking for other stuff. This is why it is crucial to pay attention to your surroundings while traversing Mars. Unexpected items can include: a) rocks that are not similar to other rocks present on the Mars Yard – it may be an important ore for future use, b) signs of extra-terrestrial life (you never know – maybe you will find something), c) signs of technological civilization (either human – related or not), and d) observations of any geological activity. You should identify up to 3 objects and prepare: 1) a short description of the object and 2) an ad-hoc hypothesis about the nature of this object, 3) an image of the object, and 4) it should include a map (drawn on the digital elevation map provided to you by the Organiser) with your traverse and the location of all found objects clearly marked on your updated geological map. The description of each object should be no longer than 350 characters (with spaces) plus one annotated photograph, and up to 3 objects will be graded. Do not afraid to be funny in your descriptions – the judges will need some entertainment during grading.

The report should be no longer than 3000 characters (with spaces) and include no more than 3 figures/annotated photographs with proper descriptions (figure caption included in the character limit) + an updated geological map.

7.3.1.2 Surface Sampling Sub-Task

The main goal of this sub-task is to collect surface samples of the Martian soil that will be sent back to Earth for further analysis. Surface and Deep Sample Collection Sub-Tasks will be performed in the same time slot.

The task scenario is as follows:

1. Arrive to the sampling location
2. Take a photo before sampling is started
3. Perform sampling (regolith and rocks samples in case of surface sampling)
4. Place the collected sample/s in the onboard container
5. Measure weight of the collected sample/s
6. Take a photo after sampling is completed
7. Drive to the start/finish line

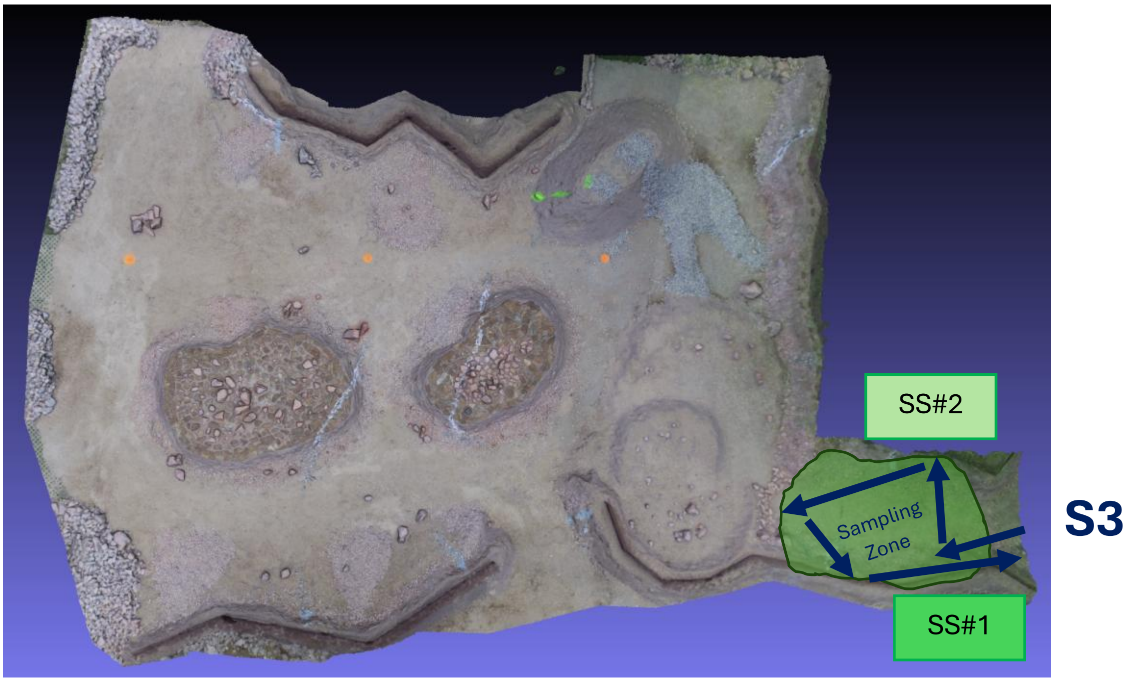

Figure 3: Example of surface sampling based on ERC 2024. Start from S3 point, rocks sampling at SS#2, regolith collection at SS#1 and return to S3.

General requirements for the surface sampling are as follows:

1. The rover shall be equipped with a manipulation device that is able to pick up the surface samples from the Mars Yard and place these in the on-board container
2. The rover shall be equipped with a container/s allowing for the stable transport of the collected material over challenging terrain
3. The container/s shall keep the collected samples in a selected position and orientation, and it shall prevent any movement
4. Regolith and rock samples (both!) must be collected by the rover. Both mass (>100g as a goal) and size (in case of rocks the largest dimension >150mm as a goal) requirements are applicable to assure that scientifically valid samples and enough material are transported back to Earth.
5. The collected samples (qty=2) must be further disposed in the onboard containers and transported to the finish line (Martian Lander Module – MLMOD).
6. Weight of the samples must be measured by the onboard system of the rover.
7. Special tool or tools must be used to perform the tasks, e.g., grippers assembled on the robotic arm.
8. Weight and dimensions of the collected sample will be checked by judges at the finish line.

SURFACE SAMPLES COLLECTION PART – GENERAL COMMENTS:

- Surface and Deep Sampling Sub-Tasks will be performed in a single run and sequence (one after another, sequence to be selected by teams)
- Sampling zones will be predefined by judges.
- Remember that in the collection part both surface and deep samples shall be collected!
- Remember that in the surface collection part both regolith and rock samples shall be collected!
- Remember about taking pictures before and after collecting samples
- Measure weight of the collected sample/s.
- Science **Surface and Deep Sample Collection Sub-Tasks** are designed to solely check rover's ability to perform collection of various samples. These samples **are not part of the Scientific Exploration activities (shall not be included in the report as the sampling location is defined by the organizer)**. In case your team want to include sample collection and results of in-situ analysis performed by the rover it has to be performed again during the Scientific Exploration Sub-Task.

7.3.1.3 Deep Sampling Sub-Task

The main goal of this sub-task is to collect surface samples of the Martian soil that will be sent back to Earth for further analysis. Surface and Deep Sample Collection Sub-Tasks will be performed in the same time slot.

The task scenario is as follows:

1. Arrive to the sampling location
2. Take a photo before sampling is started
3. Perform sampling (drilled-out core in case of deep sampling)
4. Place the collected sample/s in the onboard container
5. Measure weight of the collected sample/s
6. Take a photo after sampling is completed
7. Drive to the start/finish line

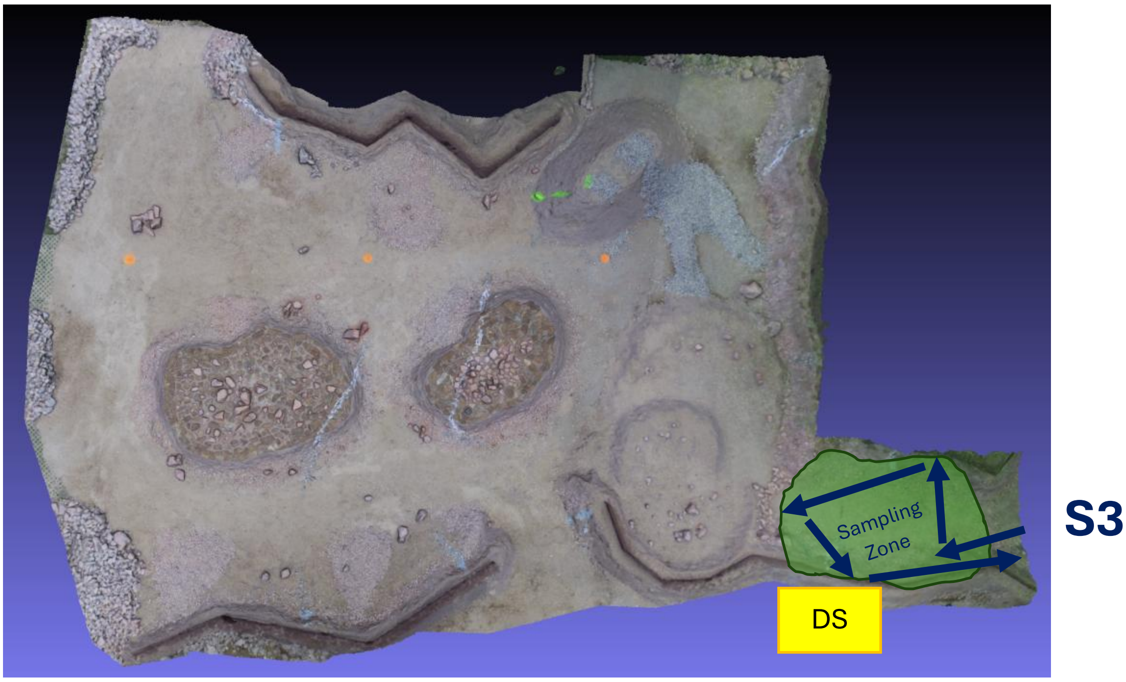

Figure 4: Example of deep sampling based on ERC 2024. Start from S3 point, deep sampling at DS and return to S3.

General requirements for the deep sampling are as follows:

1. The rover shall be equipped with a deep sample collection device that is able to pick up the surface sample from up to 30cm below the Mars Yard's surface and place it in the on-board container
2. The rover shall be equipped with a container/s allowing for the stable transport of the collected material over challenging terrain
3. The container/s shall keep the collected samples in a selected position and orientation and it shall prevent any movement
4. Deep sample must be collected by the rover. Both mass (>100g as a goal) and depth (>300mm as a goal) requirements are applicable to assure that scientifically valid samples and enough material are transported back to Earth.
5. The collected material must be further disposed in the onboard container and transported to the finish line (Martian Lander Module – MLMOD).
6. Weight of the sample must be measured by the onboard system of the rover.
7. Special tool or tools must be used to perform the tasks, e.g., drilling mechanism assembled on the robotic arm (any workarounds, e.g., drilling using rover's wheels) are prohibited.
8. Weight of the collected sample may be checked by judges at the finish line. Depth will be assessed based on the coloured layers of material in the drilling location. An example of drilling site is shown in the figure below. Moreover:
    a. Max depth is not and will not be defined. Your task is to drill to at least 30cm. After that you can expect up to a few cm of soft sand/soil
    b. There will be a coloured layer of soil at each scored level
    c. Colour codes are provided for judges to assess how deep did you manage to drill and if you really drilled to that depth instead of drilling to the side

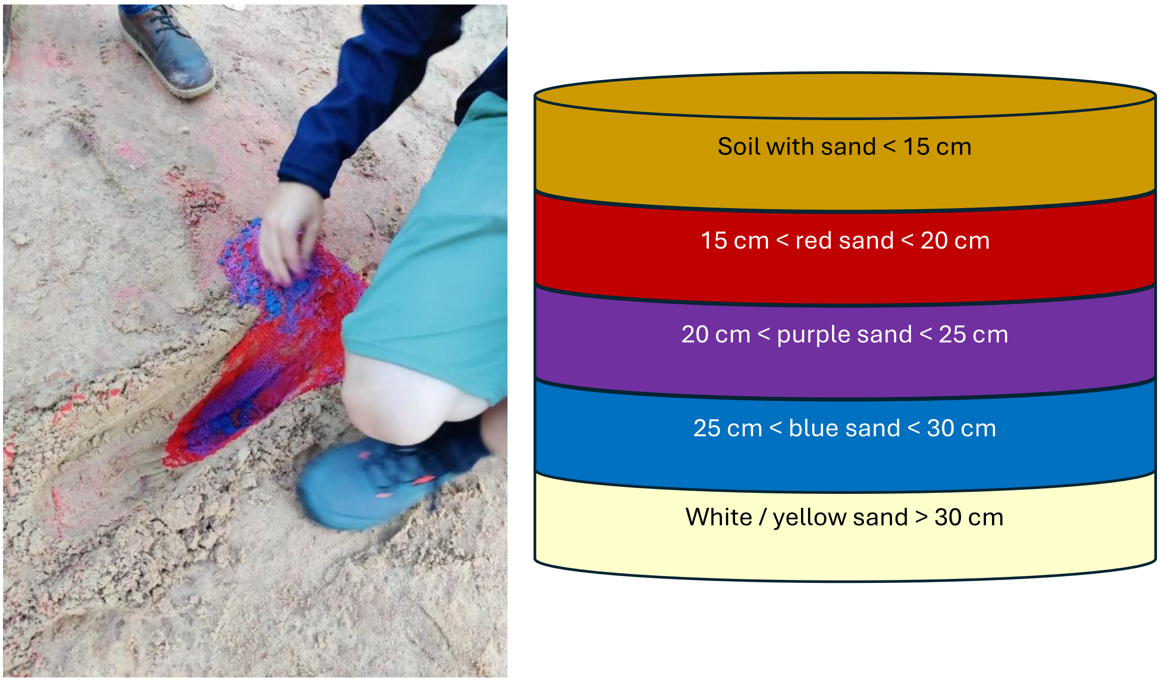

Figure 5: Example of drilling location (left) and foreseen colours of the layers for the Deep Sampling Sub-Task (right)

DEEP SAMPLES COLLECTION PART – GENERAL COMMENTS:

- Surface and Deep Sampling Sub-Tasks will be performed in a single run and sequence (one after another, sequence to be selected by teams)
- Sampling zones will be predefined by judges.
- Remember that in the collection part both surface and deep samples shall be collected!
- Remember that in the deep sample collection part deep coloured sand samples shall be collected!
- Remember about taking pictures before and after collecting samples
- Measure weight of the collected sample/s. Depth from which the deep sample was collected has to be measured too
- Science **Surface and Deep Sample Collection Sub-Tasks** are designed to solely check rover's ability to perform collection of various samples. These samples **are not part of the Scientific Exploration activities (shall not be included in the report as the sampling location is defined by the organizer)**. In case your team want to include sample collection and results of in-situ analysis performed by the rover it has to be done during the Scientific Exploration Sub-Task.
- In case of deep sampling **drilling out (without placement in some kind of preliminary container but directly on the ground) and scooping the coloured layers with the robotic arm is forbidden**. In this case you would 'contaminate' the deep sampling location for the other teams with the coloured soil (it will not be possible to scoop everything). In addition, if no coloured soil would be used (in a real life) you cannot be sure that you have managed to drill to the assumed depth - it would result in low technical excellence of your mechanism. You can drill out deep sample material, keep it in the tube around the drill and pour it in different location than deep sampling location (e.g. by simply changing rotation direction of the drill). Then you will not contaminate this location, and you can scoop the material. The perfect solution considers drilling out the sample and directly placing it into container that is used to perform weight measurement onboard and then by judges when the finish line has been reached.

7.3.1.4 Astro-Bio Sub-Task

The aim of the Science-Astrobiology Task is to demonstrate the **understanding of performing scientific exploration in planetary astro-biological conditions with accordance to the standards of Planetary Protection protocols**. The workflow of this task is as follows:

1. Teams are obligated to perform the pH-meter calibration. Organisers will provide pH standardized calibration solutions which will be available up to 1h before the start of MY time. Calibration must be done before the preparation time.
2. Planning mission based on the delivered basic Mars Yard information regarding the task (map of the Mars Yard will be provided).
3. Decision where to go - decision will have to be made basing on the provided data regarding the conditions and geophysical properties that will give the biggest chances to find life***. Data sets will be handed over to the teams 15min before the start of the MY time.
4. Identification of the sample source via QR code located in the task area. QR code will contain text data (example: Site 1, Site 2, etc).
5. Performing the measurement of the pH of water taken from plastic containers (diameter 60 mm, height 75 mm) that will be half-buried within the designated area. The measurement must include performing the control measurement on the known solution brought by the team, right before the start of MY time.
6. Performing a liquid sample acquisition from and transferring it to a secure container (not more than 100ml).
7. Safe and clean retrieving of the liquid sample from the rover. Organisers will check if any leakage occurred.
8. Organisers will hand over of the envelope with the further astro-biological data from the chosen measurement site
9. Delivery of the final report from the task up to 3 hours after completion of your MY time allocated for this task. This includes analysis of the astro-biological data (mentioned in point 3 and 8) and performing Thin Layer Chromatography (TLC) on collected liquid samples.
   *** rocks QR identifications will be randomized with respect to the measurement sites within the area of the task

Materials Provided by the ERC Organizer

You will be provided with the following data based on which you will prepare your reports:

1. Detailed instruction (below)
2. At the time of the MY digital elevation map you will be informed which sector of the area will be designated for the Science-Astrobiology task. Containers with liquid of unknown pH are going to be standardized plastic cups (diameter 60 mm, height 75 mm).
3. Set of pH reference solutions for pH-meter calibration with the standardized plastic cup (the same as on MY) that will include non-corrosive solution of known pH
4. Envelope with the set of astro-biological and geophysical data for chosen measurement site (during the task).
5. Tools and reagents necessary to perform Thin Liquid Chromatography:
   a. Solvent (50ml)
   b. Pipettes (1x)
   c. Reaction chamber (1x)
   d. TLC plate (1x)
6. A non-compulsory workshop that will introduce various topics helpful to accomplish Science Tasks 5. Support of our Science Team through the dedicated contact platform

Description and Documentation of the Sub-Task:

The goal of this task is to perform an astro-biological analysis in Martian conditions and performing a liquid sample collection with accordance to the standards of Planetary Protection protocols****. Task will consist of choosing one out of five measuring sites, performing the measurement of pH of the solution located in the measuring area and acquisition of liquid sample (up to 100ml) then storing it in a secure container in/on the rover. The last part of the task will be performed outside of the Mars Yard. After the measurement attempt and retrieving the sample (judges will check the cleanness of the container), the team will get an envelope with a set of astro-biological, geophysical and biochemical data from the chosen measurement site. Based on this information, the team will have to include in the final report basic biological analysis of the virtual sample with a preliminary evaluation potential of finding life in the chosen area. Parallely, the biochemical analysis (chromatographic methods) will have to be performed. Results of the analysis of produced astro-biological data and chromatography will have to be included in the report.

<u>Part 1: Science – Choosing the Measurement Site</u>

As the team will be ready to start the trail, judges will hand over the map of Mars with five locations marked. These locations will correspond to the five measurement sites from which the team will have to pick one. All measurement sites will be in the designated area of the Mars Yard. Map will contain descriptions with a set of basic geophysical data (mean temperature, pressure, humidity etc.). using this data team chooses the preferred measuring site based on maximization of its astro-biological potential. Justification and reasoning that lead to choosing a particular site will be graded in the final report.

Part 2: Science – Identification of the Measurement Site and Performing the Measurement

1. **QR-Code Scanning**
   There will be the QR-code near to the chosen spot with plastic container. Team will need to scan a QR-code that will contain the identification number. Identification number needs to be sent to the judges who will prepare the data necessary for further part of the challenge.
2. **pH Measurement**
   Each site will resemble a rock with the container with a measuring solution. Team will need to perform a pH measurement with the method of their choosing. Justification of the choice of the measurement technique will have to be a part of the final report. Regarding the chosen method, calibration will need to be performed within the presence of the judge at the same time as the "Part 1: Science: Choosing the measurement site" Calibration solutions and reference pH-meter will be provided by ERC organizers.
3. **Liquid Sample Acquisition**
   The team needs to take a sample of the liquid from the measurement site. The volume of the acquired sample cannot exceed 100ml. The container with the sample must be designed in such a way to allow safe and leakproof retrieving of the sample and at the same time keeping it sealed, not allowing for the crosscontamination. This step will require following the safety protocols to prevent spillage, leakage and/or coss-contamiantion. Judges will check if any leakage has occurred and if the container properly isolates the sample from the outside environment. Information about following the safety protocols and effectiveness of preventing the possible cross-contamination will have to be included in the report.

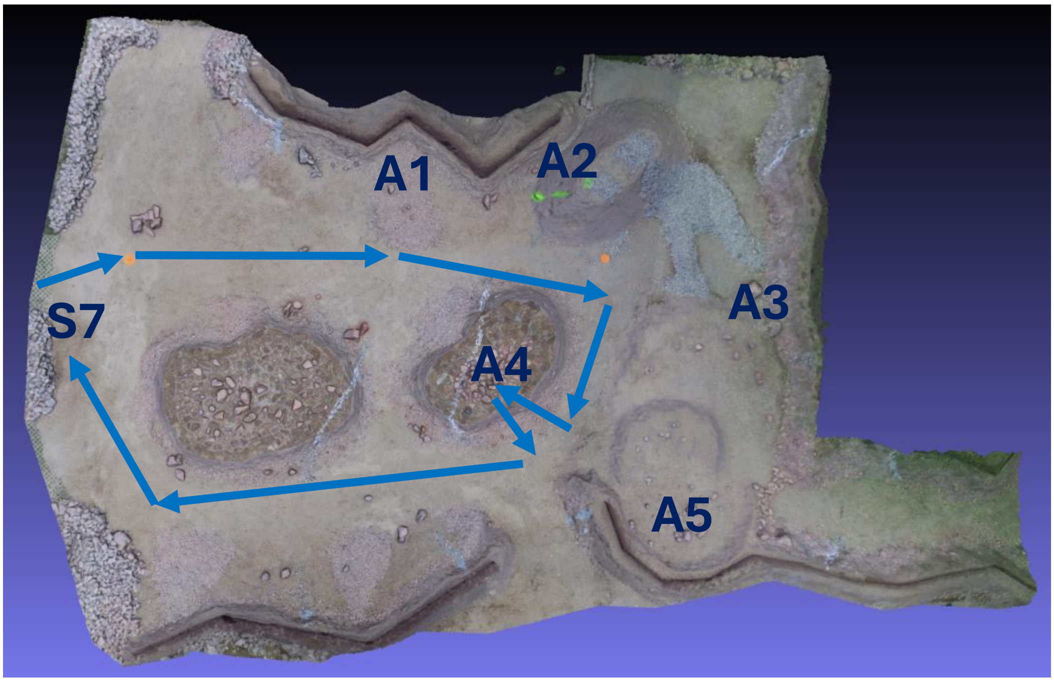

Figure 6: Example of Astro-Bio Sub-Task based on ERC 2024. Start from S7 point, selection of 1 of 5 measurement points and performance of astro-biological measurements used for data analysis. Go back to the start line. Attention! Astro-Bio Sub-Task will have a separate zone on Mars Yard in 2026 (as e.g. Sampling Sub-Tasks)

Part 3: Science – Analysis of the Astro-Biological Data

A) Data from the envelope

Based on the information from the envelope and pH-measurement team is obligated to perform basic analysis of the data and draw conclusions regarding the possibility and probability of finding life in the chosen area. Justification of these conclusions will have to consist of the reasoning behind the proper statement and scientifically accurate arguments that will support this hypothesis.

The mentioned envelope will be handed to the team after reaching the site and the attempt of performing the pH measurement. Information in the envelope will contain basic astro-biologically relevant characterisation of the site. List will contain basic geophysical and astro-biological data gathered from the measurement site. Data will include data like:

- mean temperature
- mean humidity
- mean pressure
- biochemical data (for example, presence and concentration of amino acids, DNA

B) Biochemical analysis of acquired liquid sample
Sample retrieved from the rover will have to be subjected to the biochemical analysis using chromatography. The method of choosing is Thin Layer Chromatography (TLC). All necessary tools will be provided by organisers.

The TLC kit will contain:

- Solvent (50ml)
- Pipette (1x)
- Reaction chamber (1x)
- TLC plate (1x)

Data obtained from the chromatographic analysis with description of reasoning and possible conclusions will have to be mentioned in the report. Picture of the TLC plate after Chromatography with marked significant features must be added to the report.

Part 4: Science: Preparation of Final Report

Scientific part of the final report from this challenge will have to contain several points:

- Justification and reasoning that lead to choosing a particular measurement site
- Description of chosen pH measurement method and description of the calibration of pH measuring method including the results.
- Results of the pH measurement with the results of the measurement of the control sample.
- Basic biological analysis of the virtual sample with preliminary evaluation potential of finding life in the chosen area.
- Results, conclusions and reasoning from the biochemical analysis of collected liquid sample.
- **Upper limit for this report is 1500 characters. The report shall be delivered up to 3 hours after your ride on the Mars Yard (refer to §6.4 for the delivery link).**

ASTRO-BIO SUB-TASK, GENERAL COMMENTS:

- Perform calibration of pH-meter
- Select one of 5 measurement locations. Remember that selection of this location needs to be scientifically valid i.e. you have to explain why you foresee to find life in this location.
- Perform measurements (pH and QR scan) in the selected location
- Perform the liquid sample collection in a safe and clean manner.
- Going back to the start line is obligatory.
- Remember to perform scientifically correct analysis of the virtually collected data and biochemical analysis of liquid samples in the report!
- Provide the AstroBio Final Report (length up to 1500 characters) up to 3 hours after your ride on the Mars Yard.

7.3.2 Navigation Task

7.3.2.1 Traverse Sub-Task

7.3.2.1.1 General Description

This task is intended to demonstrate the system's ability for semi to fully autonomous traversal. The team shall develop a project which gradually evolves into a fully autonomous system, traversing and gathering important data on its way. At an early stage, the system can be decoupled with the operator in the loop, but all planning and parameter estimation must be done by the computer system itself. This limits the operator to navigate the rover blindly i.e., without access to visual or any other spatial information, however, any kind of data can be processed on-board, providing the operator with support information about the localisation and state of the system. A smart navigation strategy, sensor fusion and image data processing are essential in this task.

The teams will be given **20 minutes** to perform Navigation: Traverse Sub-Task. The Teams will be given 15 minutes before the Task start to prepare the rover for the traversal. This time is called the Preparation Phase. During most of the Preparation Phase, full access to the Rover is granted, the Rover is put in a dedicated area of the Mars Yard and the Teams are allowed to drive in the area, upload software, check the functionalities, etc.

7.3.2.1.2 Task Scenario

a) Send the rover the start position and waypoint information
b) Reach the planned waypoints (4) according to the Navigation Plan
c) Return rover to its start position
d) After the traverse, present techniques used, visualise the system data, compare the results with the plan calculated at the beginning etc.

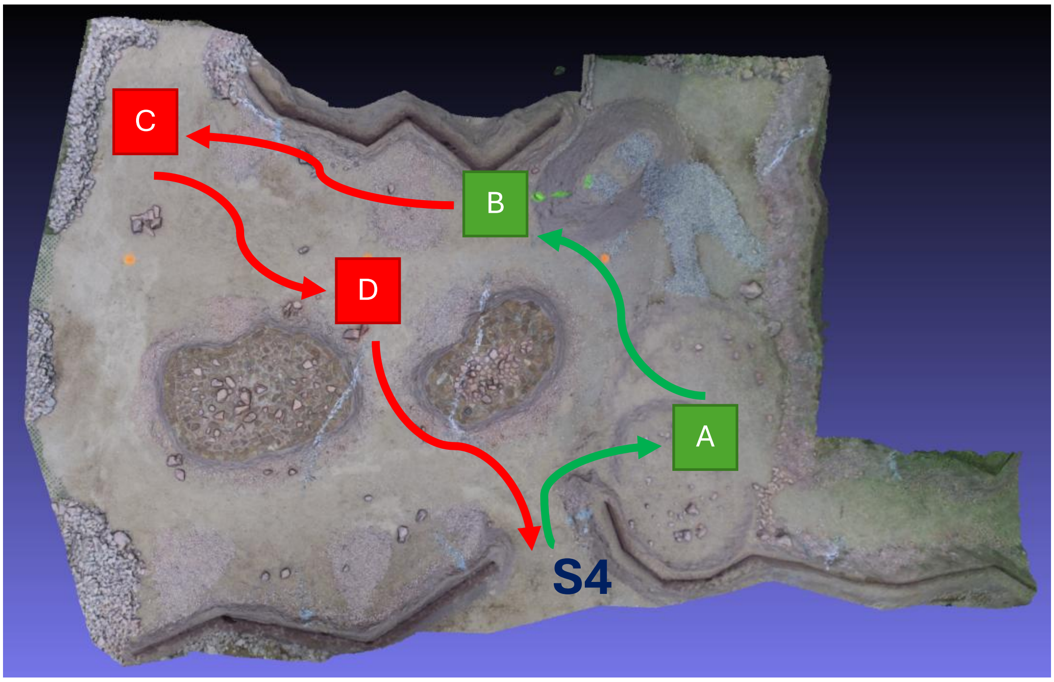

Figure 7: Example of Traverse Sub-Task based on ERC 2024. Start from S4 point, reaching points A-B-C-D and going back to the start line.

7.3.2.1.3 General Requirements

a) Reach the 4 points defined by the judges in any chosen sequence (!)

b) Reach the defined point 5 (finish) as the last one (!)

c) Perform this task without video feedback in a fully autonomous mode in order to maximise the received scores

d) Perform this task with video feedback to receive 50% of points (100% possible for autonomy if video feed-based autonomy is implemented and presented to judges – each team is responsible for presenting proof of autonomy to judges)

e) GNSS navigation can be used (assumes existence of satellite constellation over Mars)! Scoring is reduced to 80% in this case (100% possible for autonomy if GNSS-based autonomy or combination of GNSS and landmarks-based autonomy is implemented and presented to judges – each team is responsible for presenting proof of autonomy to judges)

f) Summarising the abovementioned scoring:

Case Description	Traverse Points Possible	Autonomy Points Possible
1. Video feed used **2. No autonomy implemented**	50%	0%
1. Video feed used **2. Autonomy implemented**	50%	100%
1. Video feed used **2. GNSS used** **3. No autonomy implemented**	50%	0%
1. Video feed used **2. GNSS used** **3. Autonomy implemented**	50%	100%
1. GNSS used **2. Video feed not used** **3. No autonomy implemented**	80%	0%
1. GNSS used **2. Video feed not used** **3. Autonomy implemented**	80%	100%
1. GNSS not used **2. Video feed not used** **3. No autonomy implemented**	100%	0%
1. GNSS not used **2. Video feed not used** **3. Autonomy implemented**	100%	100%

Table 3: ERC 2026 Navigation Traverse Factors

1. It does not matter what method do you use for autonomy - either you have it (=100%) or you do not have it (=0%).
2. E.g. 'Video feed used' and 'GNSS used' means that you see video feed or current location of the rover on the screen IN THE CONTROL STATION and based on this you perform rover's manoeuvres by the operator.
3. In case you do not use this information in the control station, but the rover does it onboard in autonomous mode (=WITHOUT SENDING IT TO THE CONTROL STATION / OPERATOR) you will get 100% of points for autonomy (if it really works).

Please be aware that both video feed and GNSS will not make autonomy implementation easy. You will still have a lot of features (e.g. rocks, trenches, bumps, landmarks...) in the Mars Yard that will not be visible in the 3D model. So going directly from Location #1 to Location #2 (assuming e.g. GNSS only) still includes risk of the rover flipping over or being stuck.

g) ArUco markers placed on the MY can be used to navigate the rover

7.3.2.1.4 Additional Information

a) Initial rover position and orientation will be drawn at the beginning of each trial from a set of designated locations and with a limited heading
b) The rover can be tele-operated, but only using the position and orientation estimate available. This data can be visualised in any form (e.g. projecting the rover's position on the arena map provided or a top view picture etc.)
c) If the rover needs to be moved for any reason, it can only be moved back to the last successfully reached waypoint and/or rotated towards any point or back to the starting point. In both cases, penalties for manual intervention apply
d) Technical Reports should include a list of all sensors, together with detailed information about working modes, ways they are used in the navigation task and how the rover will be operated. Teams are entitled to consult all solutions with the judges before submission. Documentation will be verified by the judges and in case of any doubts, the Team may be asked to reconfigure devices and/or their control strategy.

Any difference between the approved configuration and the one used during the challenge can lead to disqualification (0 points for this task)

Task arena:

- Final map with grid coordinates and POIs (Point Of Interest) will be provided on the warm-up day, if not agreed otherwise
- Two types of landmarks are foreseen: natural landmarks which are elements of the landscape placed on the map, e.g. craters, small embankments, hills and artificial landmarks, e.g. artificial points for localisation purposes. Artificial landmarks can contain characteristic hi-visibility labels, unique geometric figures, alphanumeric signs or an AR/QR tag matching a POI label on the map; An example landmark (A4 size) design using an ArUco marker is presented below:

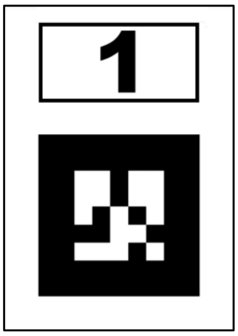

Figure 8: Example of marker

- The artificial landmarks will be visible for cameras from different directions on a field (each landmark has 4 sides with identical faces) and will have a physical base that can be detected by proximity/range sensors (e.g. placed on an element of infrastructure or natural landmark)
- At least two landmarks will be visible from the starting point
- It must be taken into account that some of the landmarks may be obscured by terrain or other objects during a traverse
- The team cannot place any additional passive landmarks or active beacons on the challenge field outside the starting area, but such elements can be deployed using the rover during the trial. All landmarks must be documented in the technical reports and presented for approval at least 10 working days before the submission of the final documentation. This equipment may be subject to negotiation, so teams should leave enough time to redesign/modify it in case of comments/rejection by the jury. Such equipment must comply with the other rules of the competition e.g. if active radio beacons are used, they must be compliant with the radio communication rules (see *Radio Communication* section) and described in RF form

e) The rover can be stopped and moved/rotated by team members when it is stuck or in case of any other technical problems. A judge shall be informed before any action is undertaken.
f) Details of the task, such as exact appearance of landmarks, location, map format, allowed custom landmarks and beacon types, etc. will be discussed with the teams and presented during the preliminary design phase in the rules. Teams are encouraged to initiate and actively participate in these discussions
g) Description of Landmarks – dimensions of landmarks are shown in the figure below.

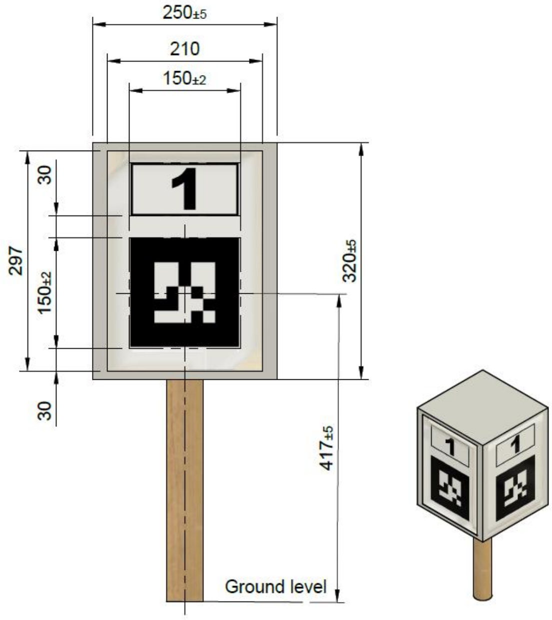

Figure 9: Dimensions of landmarks

Each artificial landmark face consists of:

- A white background of at least 210x297mm (A4 size)
- A landmark number indicator located in a 150x70mm (+-2mm) rectangle
- A marker square corresponding to the landmark number, with dimensions of 150x150mm (+-2mm) and at least 30mm white margin on each side, grid size of the tag is 5x5

Each artificial landmark pole consists of:

- A box with 4 landmark faces and dimensions of 250x250x310mm (+-5mm)
- A wooden pole that elevates the landmark box above the ground (according to the drawing)

Each landmark's coordinates are counted from an artificial point located in the landmark's pole axis – on a height of the landmark ARTag's centre point.

For the **list of ArUco markers** – please refer to the following generator.

The **size 5x5 library ArUco markers (not original library)** will be used (the final list to be confirmed in the *Update Report #3*). Markers ID No. 51-64 from the abovementioned generator are assumed at the moment.

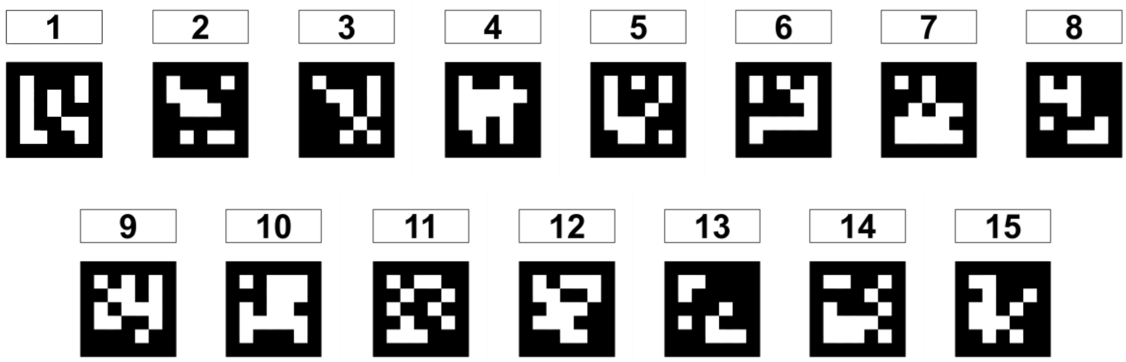

Figure 10: ArUco markers to be used on landmarks

7.3.2.2 Droning Sub-Task

7.3.2.2.1 General Description

This task is intended to demonstrate the drone's ability to perform automated mission with focus on reliable and repeatable performance.

Additionally, before performing the mission the drone will go through a preflight check to test the implementation of crucial safety features described in a later section. The drone must pass all of the checks to be allowed to perform the task. To ensure the procedure goes smoothly teams should familiarize themselves with description of the procedure provided in Appendix 6 – Preflight Check Description.

The mission consists of lift-off, detecting the landing spot and precision landing at the landing spot. The mission is to be repeated 3 times. Each time the position of the landing spot will be randomized within 3 meters radius around the lift-off spot.

Teams can receive additional points for detecting and estimating positions of 3 probes scattered within the 3 meters radius around the lift-off spot (5 points per detection in each of the 3 missions for a total of 45 points). The position of the probes will remain unchanged between the 3 missions, but detections in each of the missions are scored individually. For details regarding the probes refer to Probing Task description.

Teams can also receive additional points by preparing and using their own landing platform that can be attached to their rover. The platform doesn't have to be attached to the rover when performing this task, but teams must prove that the platform can be attached to the rover. The platform can't have a surface larger than a disc of radius 0.5 m. The platform does not have to be using an ArUco marker. Teams can receive 15 points for preparing their own landing platform and 15 points for each successful landing on it (up to a total of 45 points).

Teams must also submit a short Droning Sub-Task Report outlining solutions implemented in their drone. Details of the contents of the report are described inside the template, which can be found alongside templates of other reports. The report should be no longer than 2 pages of text (excluding images etc.). The reports shall be delivered as mentioned in the section §6.4.

Image processing / object detection and high-level logic / decision-making are required to be implemented using MATLAB/Simulink (software will be available for the teams). This task is governed by ONT (Oprogramowanie Naukowo-Techniczne) – partner of the ERC and main distributor of MathWorks products in Poland. ONT plans to organize webinars for teams in order to introduce them to software and toolboxes that shall be used for this task. They will also serve as technical point of contact (mainly on the ERC's Slack).

7.3.2.2.2 Preflight Check of Safety Features

During the preflight check teams must present the implementation of the following features, in order:

- Operator must have connection and real time feedback from the drone, with view on the critical telemetry flight parameters as well as video feed
- Drone operator must have the ability to switch to manual / remote control mode on demand (requested by Jury)
- Drone must be able to automatically maintain stability mid-air
- Drone operator must have the ability to initialize automatic landing on demand (whenever requested by Jury)
- Drone must have implemented fail safe mechanisms in case of:
  - ✓ Losing connection with remote control and ground station
  - ✓ Battery/power loss, positioning system glitch
- Operator must set the polygon inclusion geofencing for the competition area The geofencing area doesn't have to be precise. Make it big enough to avoid false triggers. The most important is to show that you are able to set geofencing for your drone.

7.3.2.2.3 Additional Information

- Drones must be assembled by teams (no COTS drones allowed).
- Each team has **15 minutes to prepare** to the task (outside the cage) and **30 minutes to execute** the task.
- Mission is to be performed **only** in automated mode. Any manual interventions during the flight will be penalized.
- After landing the drone will be manually placed back at the lift-off spot.
- The size of the cage is 10 x 10 x 4 m.
- The effective area of competition is a disc with a radius of 3 m in the middle of the cage.
- The lift-off spot is a 1 x 1 m square in the middle of the cage.
- The landing target provided by the organizers is a disc with a radius of 0.5 m.
- Teams may also prepare their own landing targets attachable to their rover, but its size must be confined within a disc with a radius of 0.5 m.
- The lift-off spot and the landing target will be marked with an ArUco marker from the **ArUco original library.** The lift-off spot will us marker ID 101 and the landing spot will use marker ID 102. The markers size will be 15 x 15 cm.
- The landing is considered as successful if any part of the drone is touching the landing target.
- The area of competition will be virtually divided into 1 x 1 m sectors. The probe detection system must point to locations of the probes using IDs of those sectors as shown on the diagram below. As an example, sector A2 covers coordinates (0 m; 0.5 m) to (0.5 m; 1 m), D1 covers (-1.5 m; 0 m) to (-1 m; 0.5 m) etc.

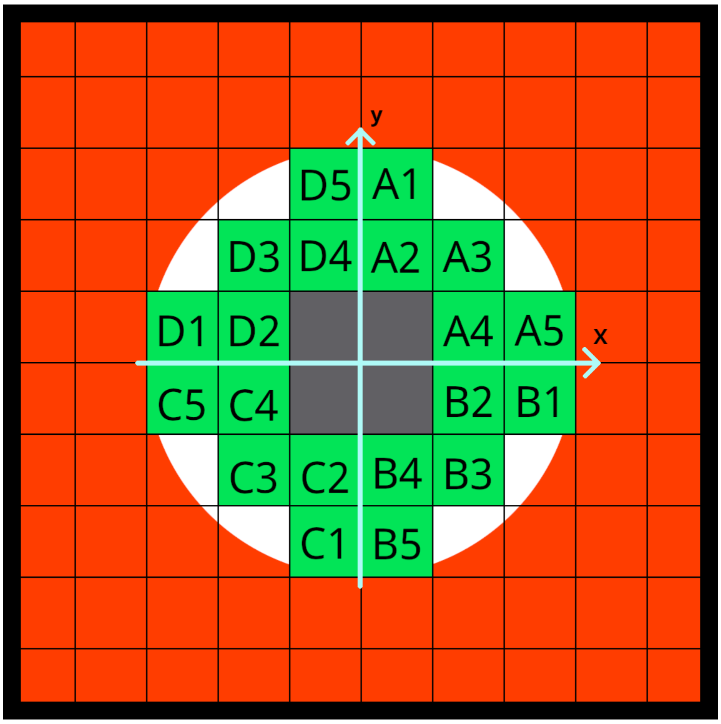

Figure 11: ERC 2026 Droning Sub-Task – Sectors

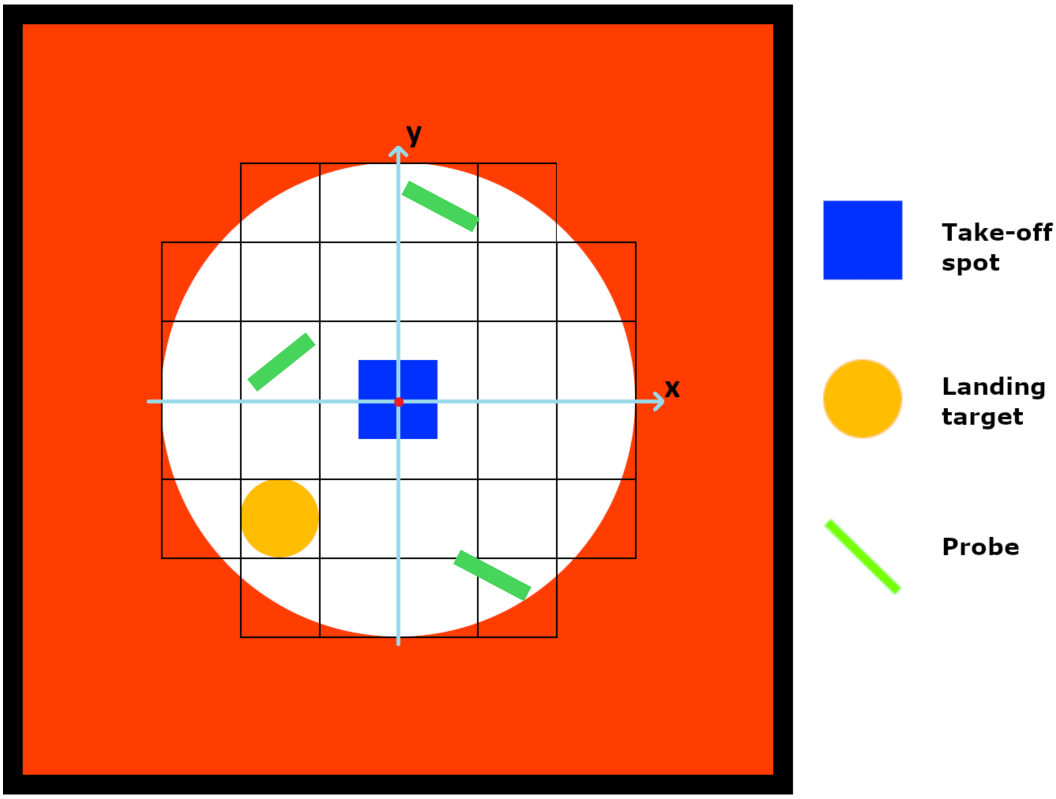

Figure 12: ERC 2026 Droning Sub-Task plan

7.3.3 Maintenance Task

7.3.3.1 General Description

The task is intended to demonstrate the ability of rovers to interact with a panel primarily designed for human use. The Team shall use the rover's robotic arm to set or rotate switches to required positions, handle different kinds of elements (light plastic plug or heavy metal bar) and gather important feedback.

7.3.3.2 Technology Priorities

a) Task automation

- Automatic element detection (spatial parameters, possible actions)
- Automatic rover or end-effector approach
- Automatic handling

b) Teleoperation interface

- Situational awareness (vision, parameters presentation, display ergonomics)
- Feedback (presentation of feedback measures, force-feedback/control interfaces)
- Ergonomics of the operator's interface

c) Robotic arm performance

- Tool's relevance for a specific operation
- Operation robustness (flexibility, dexterity)
- Operation accuracy and quality

7.3.3.3 Task Scenario (Operations)

1. Set the main switch to the ON position as the first operation.
2. Set four switches from the group of lever switches.
3. Turn three of the five rotary switches.
4. Insert the plug into one of the two sockets (IEC320 C14 type).
5. Turn both rotary power switches (the first one is to power the cable connection, and the second one is to power the electromagnet).

6. Place the plate on the electromagnet.

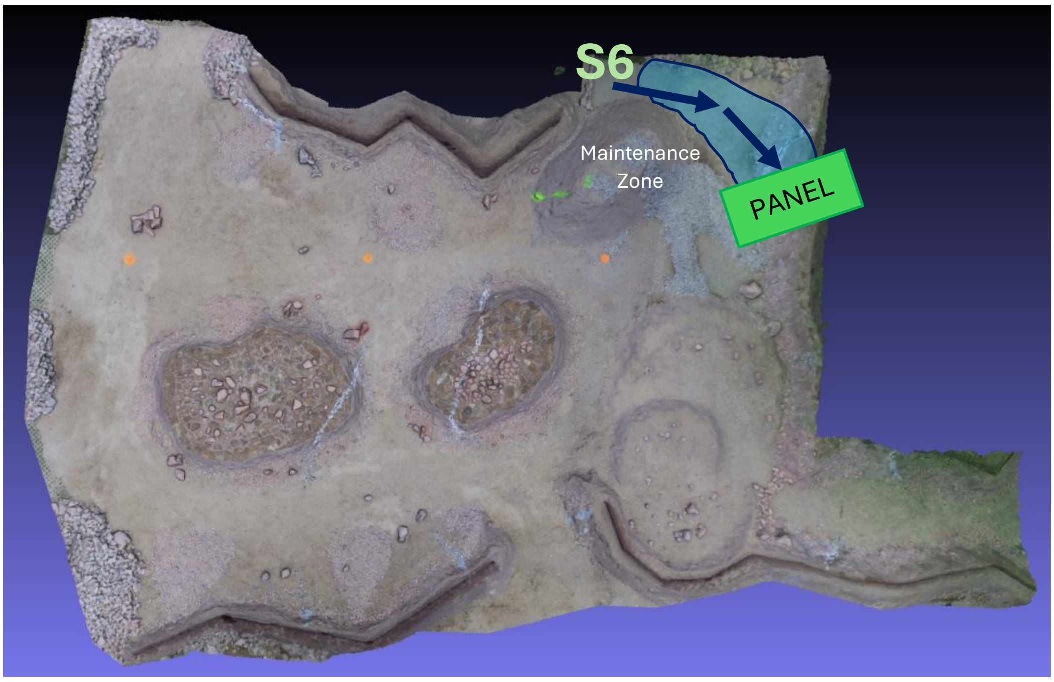

Figure 13: Example of Maintenance Task based on ERC 2024. Start from S6 point, reaching the panel and perform manipulation sequence.

7.3.3.4 Task Details

- The time in the field for this task is limited to 30 minutes.
- At least 15 minutes of preparation time is given prior to execution of the task.
- The workplace will be located at an angled side of a lander's module.
- At the workplace there will be a flat panel, 330mm wide and 450mm tall, divided into sections with specific elements (ArUco tags, indicators, switches, sockets).
- Switches and other controls on the panel will be industrial grade elements. Switches can be either lever or rotational type.
- All switches should be manipulated one by one - actuation of more than one in a single move will not be scored.
- Some of the panel elements can be sensitive to forces and torques exceeding operational limits. Those elements should not be damaged during operations.
- Dimensions provided in the description may slightly change.

7.3.3.5 Expected Results

- The MAIN switch is set to the ON position as the first operation (panel powered up).
- Four other switches are set to required positions.
- Three rotary switches are turned to required positions.
- Plug is inserted into one of the two sockets (cable connection established).
- The first rotary power switch is turned to the ON position (current for the cable connection supplied).
- The second rotary power switch is turned to the ON position (electromagnet powered up).
- The plate is held by the electromagnet.
- All of the five indicators are lit up.
- No panel damage is incurred (control elements, connectors, covers etc.).
- Task automation efforts and results are presented to the judge.

7.3.3.6 Additional Information

- Remember about the required weight and radio check before coming to the Maintenance starting point! Consider additional 20-30 minutes before, as checking will have to be done in the Team Zone before arrival to the Mars Yard.

- Arrive early, you will have 15 minutes to prepare for the task (during this time you have to establish communication with the rover and place it on the start line).
- Judge will explain you the task during the preparation phase.
- You have 30 minutes to complete the task.
- Hardware changes/repairs done on the rover during the task will be penalised. No penalties will be given for the safety actions, e.g. when rover is going to flip over on the crater's edge and you need to catch it in order to avoid damages.
- Remember about preparing the strategy for each task. What to do in case of any problems? How long should we try to perform certain action before giving up and moving to the next one? Planning in advance will save you time and allow to score more efficiently.

Figure 14: ERC 2026 Maintenance Task panel. The front workspace panel is 330 mm wide and 450 mm tall. Note: the placement of specific elements is only illustratory here and is subject to change.

7.3.4 Probing Task

7.3.4.1 General Description

The task is intended to demonstrate the ability of rovers to perform fast samples or probes collection for scientific reasons and delivery of the collected equipment to the module that has to deliver those back to Earth for examination. It is based on e.g. MSR – Mars Sample Return (NASA, mission that was assumed to go to Mars and collect samples from NASA's Perseverance rover) or SFR – Sample Fetch Rover (ESA).

7.3.4.2 Task Scenario

a) Start driving from 1 of 6 starting locations.
b) Find three probes somewhere on the Mars Yard. Store each probe on the rover in its onboard container.
c) Go to the finish line when 3 probes are collected.

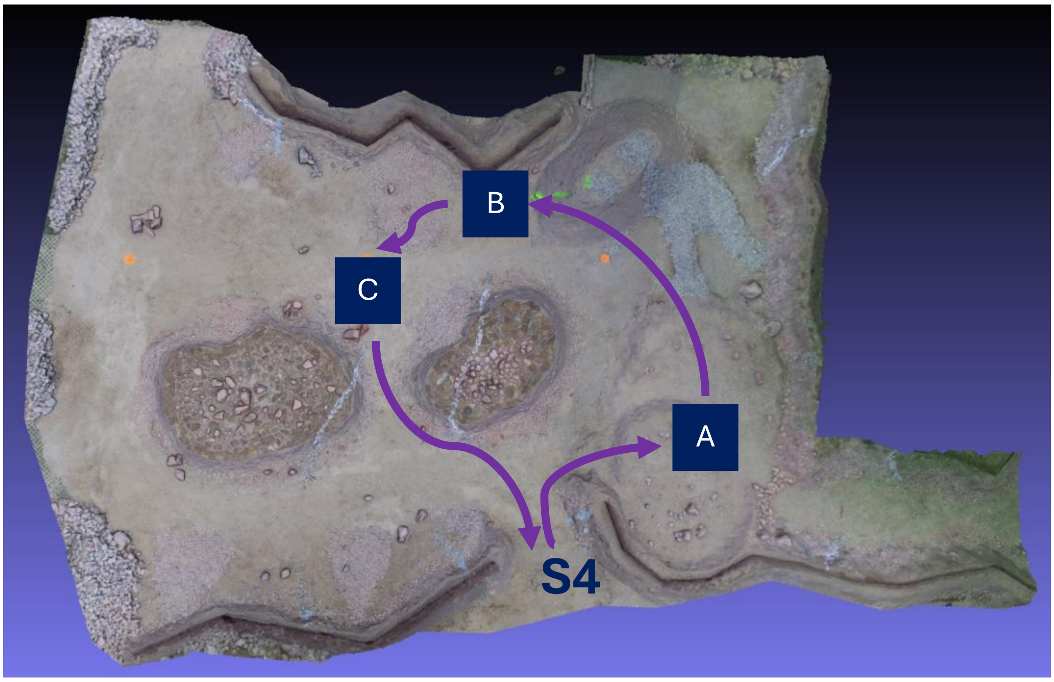

Figure 15: Example of Probing Task based on ERC 2024. Start from S4 point (or another location), collecting 3 probes and going back to the start line.

7.3.4.3 General Requirements

a) Colour of the probes: yellow-green

b) Probes are made of aluminium. Dimensions of the probes are provided below

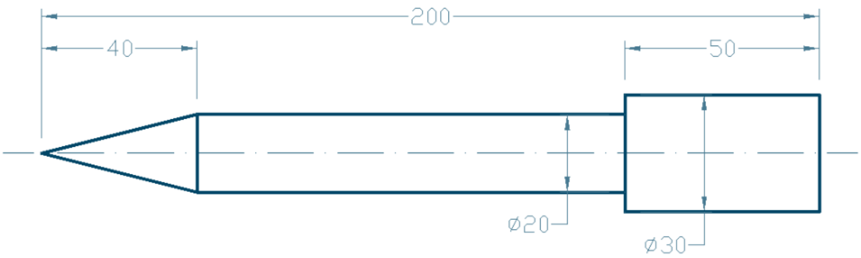

Figure 16: Dimensions of the probes

7.3.4.4 Additional Information

a) Perform this task as fast as possible. The final scores will depend on the task's completion time as follows

- Teams that complete this task in less than 7:30 minutes get 100% of the collected points.
- Teams that complete this task in more than 7:30 and less than 12:30 minutes get 90% of the collected points.
- Teams that complete this task in more than 12:30 and less than 17:30 minutes get 80% of the collected points.
- Teams that complete this task in more than 17:30 and less than 22:30 minutes get 70% of the collected points.
- Teams that complete this task in more than 22:30 minutes get 60% of the collected points.

b) Guidance on collisions avoidance and 'ownership' of the probes will be provided before the competition (Briefing for the Teams). As a baseline judges will have the final decision on which team gets to the certain probe first.

7.3.5 Presentation Task

7.3.5.1 General Description

The Presentation Task lets teams introduce themselves and present their projects. Judges expect to learn how the team worked on the project, what kind of technical solutions are implemented in the rover or in the software, what was the approach of the team to solve tasks during the competition (e.g. electro-mechanical design, algorithms), and how the team solved problems and issues which occurred during development (lessons learnt). The team should also be prepared for a Q&A session and discussions with judges. The last part is also important for the teams as the organizer would like to focus more on mentoring and development of the team members. Therefore, we would like to encourage you to prepare not only for the presentation part but also for the discussion. Questions from the team members are more than welcome!

7.3.5.2 Goals

a) Introduce the team (expertise and experience) and project
b) Present organisational structure, management methods and work-flow
c) Present an engineering approach
d) Present the technical design
e) Present the scientific project design
f) Present the approach selected and realised in each task
g) Present difficulties which occurred, methods applied to solve them as well as lessons learnt
h) Present elements designed to fulfil the rest of the trial tasks
i) Present Team's outreach and PR activities related to the ERC (posts, events etc.)
j) Present well-considered and realistic ideas for potential applications and future development of the designed rover or its subsystems (spin-off/spin-out ideas)

7.3.5.3 General Requirements

a) The time for the presentation is limited to 20 minutes
b) The Q&A and discussion session takes up to 20 minutes
c) The team can use a projector provided by the Organiser (HDMI connector as a standard, other connectors may not be available therefore please consider bringing those as an essential to present)
d) The Organiser does not provide computers
e) The presentation as well as Q&A and discussion session must be given in English
f) The presentation can be done in any format and creativity is welcome
g) One member of the Team may be selected to present, however, all Team Members shall attend the presentation as well as the Q&A and discussion sessions

7.3.5.4 Expected Results

a) Demonstration of Team's presentation skills
b) Detailed information on technical key drivers which influenced the Team to build this design, engineering approach, system breakdown structure, management, difficulties, solutions and lessons learnt
c) Scientific/engineering inventions, design propositions
d) Spin-off, spin-out/in ideas and opportunities
e) Team outreach and promotion of the ERC
    - Discussion with Judges focused on the presented design or approach, problems encountered, possibilities to further develop the project or certain skills of the Team Members
    - Teams should feel more than welcome to provide additional topics that they would like to discuss. One of the main assumptions of this task is to provide as much mentoring support to the Teams during the ERC finals.

PRESENTATION TASK – GENERAL COMMENTS:

- Arrive on time, you will have **up to 5 minutes to prepare** for the task and solve the potential technical issues related to sharing the screen or the Internet connection

- Do not select only one representative for the Presentation Task. We want to meet and talk to the team (or at least as many members as it is possible)

- Do not underappreciate this task. Presentation skills will be essential in your professional career. Prepare for this task in advance

- Judges will explain you the task during the preparation phase

- You have **20 minutes to complete the presentation**

- In the presentation please especially remember about addressing the following topics (we have noticed in the recent editions of the ERC that teams forget about those or do not have idea what to present):
  - ✓ Explanation of outreach part. Describe the activities that increase your and ERC's visibility (e.g. posts on Facebook, articles in newspapers, events/workshops that you have organised etc.). Also include PR activities related to the ERC (posts, events etc. – *at least 5 posts including **#ERC2026** hashtag will be required and scored*)
  - ✓ Explanation of spin-off/out part. Come up with ideas for further commercialisation of your work. Try to think outside the box and consider more specific possibilities, e.g. commercialisation of a small part (e.g. software, gripper, camera etc.) of the rover instead of the complete rover (however it is also a possibility but not the only one)

- Use the **20-minutes slot for the discussion with the judges**. Do not be afraid, this time is specially for you than for us. You can discuss any topic with us (considering that we have some knowledge in a certain area, if not we will try to help you and lead you to the person who has experience in it). Do not hesitate to ask any questions, even the stupid ones

7.4 Time Available for Tasks and Locations

The table below summarizes time available for preparation and completion of each task. Preparation time is granted to each team before starting the task. The table below also provides locations of the tasks and the number of teams that compete in this task at the same time (e.g. need to share Mars Yard and avoid collisions with other teams).

<table>
<tr><th>No.</th><th>Task: Sub-Task</th><th>Preparation time [min]</th><th>Execution time [min]</th><th>Location</th><th>Max number of teams competing at the same time</th></tr>
<tr><td>1.1</td><td>Science: Exploration</td><td>15</td><td>20</td><td>Mars Yard – Main Zone</td><td>3</td></tr>
<tr><td>1.2</td><td>Science: Surface Sampling</td><td rowspan="2">15</td><td rowspan="2">30</td><td rowspan="2">Mars Yard – Sampling Zone</td><td rowspan="2">1</td></tr>
<tr><td>1.3</td><td>Science: Deep Sampling</td></tr>
<tr><td>1.4</td><td>Science: Astro-Bio</td><td>15</td><td>20</td><td>Mars Yard – Astro-Bio Zone</td><td>1</td></tr>
<tr><td>2.1</td><td>Navigation: Traverse</td><td>15</td><td>20</td><td>Mars Yard – Main Zone</td><td>1</td></tr>
<tr><td>2.2</td><td>Navigation: Droning</td><td>15</td><td>30</td><td>Drone Cage (not in the Mars Yard)</td><td>1</td></tr>
<tr><td>3.1</td><td>Maintenance</td><td>15</td><td>30</td><td>Mars Yard – Maintenance Zone</td><td>1</td></tr>
<tr><td>4.1</td><td>Probing</td><td>15</td><td>30</td><td>Mars Yard – Main Zone</td><td>6</td></tr>
<tr><td>5.1</td><td>Presentation</td><td>5</td><td>20 (presentation)
20 (discussion)</td><td>Presentation Room (2 rooms are foreseen)</td><td>2
(but in different presentation rooms)</td></tr>
</table>

Table 4: Time allocations, locations and number of teams competing in each task

Appendix 1 – Schedule

The table below presents the schedule of the ERC 2026. If not stated otherwise, the below dates mean end of day (23:59) Warsaw, Poland time (UTC+2 / CEST time starting on 29^{th} March 2026).

Event	Date
Rules Publication / Registration Start	20^{th} February 2026
Registration End Deliverables: Registration Form.	15^{th} March 2026
Preliminary Design Review (PDR) Deliverables: ▪ *Preliminary Report, including e.g.* 1. *Updated MoC / Verification Control Document (VCD)* 2. *Preliminary Test Plan (TP)*	8^{th} April 2026
Video Material Delivery	31^{st} May 2026
Qualification Announcement	21^{st} June 2026
Critical Design Review (CDR) Deliverables: ▪ *Final Report, including e.g.* 1. *Final MoC / VCD* 2. *Final TP* 3. *RFF (Radio Frequency Form)*	26^{th} July 2026
Update Report #3 (incl. details about Mars Yard)	13^{th} August 2026
Exploration Task Readiness Review (E-TRR) Deliverables: ▪ *Science Planning Report*	23^{rd} August 2026
Competitions - ERC final event	**4^{th} – 6^{th} September 2026**
Registration and warm up day for teams	*3^{rd} September 2026*
Competitions – ERC tasks	*4^{th} – 6^{th} September 2026*
Closing ceremony	*6^{th} September 2026*

Table 5: ERC 2026 Schedule

Event	Date
Q&A Session #1	22^{nd} March 2026
Q&A Session #2	15^{th} April 2026
Qualification Announcement	28^{th} June 2026
Q&A Session #3	19^{th} July 2026
Q&A Session #4 (pre-ERC briefing)	3^{rd} September 2026 (face to face in Krakow)

Table 6: Q&A Sessions Schedule

Appendix 2 – Scoring

The scoring of the competition is designed to reflect the technology priorities identified as well as gaps in the designs from previous editions. It is intended to motivate teams and, as a result, help the team develop the relevant expertise and provide missing solutions to previous challenges.

Some flexibility in scoring is left to reflect many aspects that cannot be covered by rules like the quality of solutions, its robustness and performance, but judges will be equipped with detailed guidelines to make the scoring as objective as possible.

The following scoring for tasks is considered:

- **Documentation (400 points max), including:**
    1. Preliminary Report (150 points max)
    2. Video Material (100 points max)
    3. Final Report (150 points max)
- **Science Task (1000 points max), including:**
    1. Exploration (300 points max)
    2. Surface Sampling (200 points max)
    3. Deep Sampling (200 points max)
    4. Astro-Bio (300 points max)
- **Navigation Task (600 points max), including:**
    1. Traverse Task (300 points max)
    2. Droning Task (300 points max)
- **Maintenance Task (300 points max)**
- **Probing Task (200 points max)**
- **Presentation Task (300 points max)**
- **Rover's and Drone's Weight (200 points max, 6x30 + 1x20 pts)***

***(6+1) rover + drone rides/flights in total includes: Exploration (30 pts), Surface and Deep Sampling (30 pts – both are performed in a single run), Astro-Bio (30 pts), Traverse (30 pts), Droning (20pts), Maintenance (30 pts), Probing (30 pts).**

The total number of points in the ERC on-site formula is 3000.
See the attached file: **Appendix_2-Scoring**.

Appendix 3 – Requirements

See the attached file: **[Appendix 3] Requirements Rev.1.xlsx**.

Please fill the table with details relevant for your rover as it is explained in the Preliminary Report's template.

The updated table shall be delivered in the *.xlsx format as a separate file with name the same as the PDF file with Preliminary Report or Final Report. Naming is as follows:

- *TeamName_PreRep.pdf / TeamName_PreRep.xlsx*
- *TeamName_FinRep.pdf / TeamName_FinRep.xlsx*

This attachment is not considered for the total length of the document (refer to document templates for more details on the document requirements).

Appendix 4 – RFCC

See the attached file: **[Appendix 4] RFCC Rev.1.xlsx**.

Appendix 5 – RF Form

See the attached file: **[Appendix 5] RF Form Rev.1.xlsx**.

Appendix 6 – Preflight Check Description

Before a team is allowed to attempt the Droning Sub-Task, they must successfully pass the pre-flight check of all the features listed in section "7.3.2.2.2 Preflight Check of Safety Features". This attachment contains description of standard course of actions of the pre-flight check. It is meant to serve as a guide for participating teams to prepare to ensure the check is conducted in a timely manner.

1. The drone may already be placed in the middle of the cage.
2. The team shall present their configuration of the geofence.
3. The team shall show/explain their implementation of fail-safe mechanisms.
4. The remaining features are going to be presented in action. Drone's pilot should always announce their actions ahead of time. If at any point the drone gets too close to the edge of the cage, pilot shall intervene to move the drone away from the net or attempt to perform emergency landing. While performing this test flight the team shall show Jury how they receive live telemetry readings and video feed.
5. First, drone's pilot shall manually (remotely) pick the drone off the ground and move around to demonstrate that the drone is capable of stable operation.
6. Then, on Jury's instruction, pilot shall either switch drone into automated mode or land the drone and start it up in automated mode. The automated mode may either be the program used to execute the Droning Task or a different automated flight mode prepared for the purpose of the pre-flight check.
7. On Jury's instruction, pilot shall take manual (remote) control over the drone.
8. On Jury's confirmation, pilot may land the drone, and team may prepare the drone for the task.

END OF THE DOCUMENT